

AIR QUALITY IMPACT ASSESSMENT

prepared for

AUTOX (Pty) Ltd
c/o Stanford and Lindsay Road, Korsten
GQEBERHA, EASTERN CAPE

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Table of contents

1	Introduction	3
2	Site location	5
2.1	Surrounding land use.....	6
2.2	Sensitive receptors	7
3	Process description.....	8
4	Dispersion modelling study	10
5	Model input	10
5.1	Meteorological data	10
5.2	Emissions	11
5.2.1	Particulate matter (PM) and lead (Pb).....	12
5.2.2	Sulphur trioxide and sulphuric acid (H ₂ SO ₄) mist	12
5.2.3	Sulphur dioxide and nitrogen dioxide	13
5.3	Emission rates.....	13
5.4	Receptor grid and model domain.....	15
6	Compliance framework	15
7	Results.....	17
7.1	Scenario A — site only, AEL application basis.....	18
7.2	Scenario B — site only, actual expected emissions	31
7.3	Scenario C – cumulative impact	42
7.4	Comparison of the scenarios	53
8	Impact on overall air quality.....	54
8.1	Particulate matter and lead.....	54
8.2	Sulphur dioxide and nitrogen dioxide	55
8.3	Sulphuric acid	57
9	Discussion	58
9.1	Model reliability.....	58
9.1.1	Emissions	58
9.1.2	Meteorological and terrain data	59
9.1.3	Terrain and building data	59
10	Risk Assessment	60
11	Conclusions.....	62
APPENDIX A	List of sources.....	63



1 Introduction

AutoX (Pty) Ltd (AutoX) operates a lead-acid battery manufacturing plant in Korsten, Gqeberha (Port Elizabeth) in the Nelson Mandela Bay Metropolitan Municipality, Eastern Cape. The plant manufactures automotive and industrial lead-acid batteries, encompassing lead oxide production, plate making, curing, assembly, acid filling and formation charging.

Lead-acid battery manufacturing involves processes that are included in the "*List of Activities That Result in Atmospheric Emissions*" as published in Government Notice 893 of 22 November 2013 (GN893), as amended. Subcategory 4.13: "*Lead smelting*" and Subcategory 7.2: "*Production of Acids*" are applicable to some of the processes used. AutoX thus operates under the conditions of an atmospheric emissions licence (AEL), current AEL number NMBM AEL 13/36 (02).

AutoX intends to expand the operation with an additional battery assembly line. This report presents the air quality impact assessment of the plant considering the proposed expansion. The assessment includes the existing installation together with the proposed expansion stack (AX4) and stacks currently included in the AEL but not operational yet (AX23 – AX28).

LAQS modelled the dispersion of particulate matter (PM₁₀), lead (Pb), sulphur dioxide (SO₂), nitrogen dioxide (NO₂) and sulphuric acid (H₂SO₄) mist to estimate ground-level concentrations. LAQS evaluated the estimated ground-level concentrations against the "*National Ambient Air Quality Standards*" as published in GN1210, (NAAQ standards), and, where these are not available, health-based reference values published by a recognised authority. The NAAQ standards are set at limits which protect human and environmental health.

This report further assesses the impact of the AutoX plant in a cumulative assessment, considering the plant together with the other emission sources in the surrounding airshed, where data was available.

As required by the Regulations, an air quality impact assessment assumes that emissions occur at the maximum permitted concentration. Emission rates are accordingly derived from the volumetric flow of each stack and the maximum permitted concentration of each regulated pollutant applicable to that stack, rather than from measured stack concentrations.

The emissions that the AEL schedule permits and the emissions the plant can realistically be expected to release differ significantly. LAQS therefore modelled three scenarios, which are reported separately in Section 7 and compared in Section 7.4:

Scenario A: Site only, with emissions derived from the flow rates given in the AEL application form and the maximum permitted emission concentration applicable to each stack.

Scenario B: Site only, using the emissions the plant is expected to release based on the 2025 measured emission rates and using estimated emissions for the proposed expansion based on similar process measurements.

Scenario C: Cumulative impact, using the site emissions as per Scenario B together with actual emissions from surrounding industrial sources.

Scenario A is the assessed case with respect to the allocated airspace in the licensing sense: it is the impact of the emissions that the licence would authorise. Scenario B is the impact of the



same plant based on the emissions it is expected to release, on its own. Scenario C considers the combined effect of the plant and nearby industrial sources based on actual emissions.

All three scenarios use the same meteorology and modelling methodology. They differ only in the emissions inventory.

This report discusses the dispersion modelling work and risk assessment outcome in detail.



2 Site location

The plant is located in Gqeberha at 33.9267° S, 25.5732° E at a base elevation of 46 m above mean sea level. The plant's location relative to surrounding land uses is shown in Figure 1.



Figure 1: AutoX site location. The site boundary is outlined in yellow; map extent approximately 2 km x 2 km.



2.1 Surrounding land use

The sensitive receptors and land uses surrounding the site are summarised in Table 1.

Direction from site	Land use
North	Approximately 0.9 km north of the AutoX facility is the Young Park residential area that extends to 1.6 km north of the facility. Algoa Park extends from 1.9 km north of the facility to approximately 4.7 km north of AutoX. Salt Pan extends from 4.7 km north of the site to 5 km north of the site. Additionally, the Kwaford, Kwazakhele and New Brighton residential areas are situated in a northerly direction from the AutoX facility.
East	Neave Industrial extends to 0.8 km east of the site. 1.3 km east of AutoX lies Holland Park, which is bordered by Ferguson and Sydenham residential areas at approximately 1.5 km and 2 km east, respectively, of the site. The Indian Ocean is situated 3.5 km east of the facility, while the North End industrial area lies 2.7 km southeast of AutoX.
South	Korsten residential area extends for 1.2 km south of AutoX. Approximately 1.2 km south of AutoX lies the Newton Park residential area. Newton Park extends to 4 km south of the facility, and 4.8 km south of the site lies the Walmer residential area.
West	West of the facility lies the Schauderville residential area, which extends for 1.4 km west of AutoX. The Parkside residential area is approximately 1.6 km west of the site and extends for 2.2 km west of AutoX. The Malabar residential area is 2.5 km west of AutoX, which is met by the Westering residential area approximately 3.6 km west of the site.

Table 1: Land use surrounding the site.



2.2 Sensitive receptors

The following prominent sensitive receptors were identified within a 2.5 km radius of the site, summarised in Table 2.

Receptor name	Type	Estimated distance	Direction from plant
Livingstone Hospital	Hospital	400 m	North-West
Livingstone Hospital Primary School	School	400 m	North-West
Frank Joubert Primary School	School	500 m	West / South-West
De Vos Malan Primary School	School	800 m	South / South-East
Life Mercantile Private Hospital	Hospital	1.1 km	South-East
G J Louw Primary School	School	1.2 km	South-East
Adolph Schauder Primary School	School	1.2 km	South / South-West
Abraham Levy Primary School	School	1.5 km	West
Phakamisa High School	School	1.4 km	North-North-West
St Teresa's (RC) Primary School	School	1.5 km	South
St James (RC) Secondary School	School	1.5 km	South-South-West
Dietrich Primary School	School	1.6 km	South-West
Empilweni TB Hospital	Hospital	1.7 km	North-East
Sydenham Primary School	School	1.7 km	South-East
Ntyatyambo Primary School	School	2.1 km	North-West
Emzomncane Primary School	School	2.3 km	North-West

Table 2: Sensitive receptors surrounding the site.



3 Process description

As per the revised AEL application, with the addition of BM line 3, the plant will have the capacity produce a maximum 2.6 million batteries per annum. It is expected that the realized production of the plant will be 1.6 million batteries per annum following the expansion. This involves the smelting of a maximum of 26 000 tons of lead and alloys and the consumption of approximately 10 000 m³ of diluted sulphuric acid on an annual basis.

The production of lead-acid batteries at the plant entails the following steps as outlined in the AEL application form:

Lead melting and lead oxide manufacturing:

- Lead ingots are fed via a conveyor to a melting pot.
- Pure molten lead from the melting pot is fed into a reactor along with water and ambient air.
- In the reactor, the molten lead is oxidized to form lead oxide (PbO, PbO₂, Pb₂O₃).
- The lead oxide particles are blown up through a classifier, which separates the large particles from the small particles. The large particles fall down through the classifier back into the reactor.
- Fine lead oxide particles exit the classifier and travel to a cyclone, which is followed by a bag filter.
- After the bag filter, a high-efficiency particulate arrestor (HEPA) filter separates the lead oxide particles from the air stream.
- Air that is removed from the stream exits the exhaust through the post-classifier stacks AX14 and AX15.
- Air is introduced at the blending screw conveyor to transport the lead oxide through the feed line to the storage silos.
- The transport air is then cleaned via a bag filter and exhausted through stacks AX12 and AX13.
- The lead oxide manufacturing area is served by a multi-point extraction system that feeds into stack AX10.

Plate manufacturing (positive and negative):

- In the plate strip casting process, lead ingots are melted down and cast to create lead strip coils.
- In the Plate Manufacturing Process, the battery plates are made from the lead strip coils produced in the Plate Strip Casting process.
- A lead strip coil is fed through the expander line to expand the strip into a parallel pair of continuous grids that are fed through the paster, where they are coated with Lead Oxide paste.
- The coated grids are then cut to size to form the positive and negative plates.



- In the Paste Mixing process, ambient air, water, acid, and other additives are mixed with lead oxide to form a paste.
- The paste mixer is served by a wet scrubber that leads into stack AX16.
- The plates are placed into curing ovens for processes, after which they are stored in the staging area for work in progress.
- Stack AX9 is an extraction stack for activities within the plate manufacturing process, including the molten lead pots that may release lead fumes.
- There are two types of curing ovens present at AutoX: indirectly heat gas-fired ovens and electric ovens.
- Curing occurs between 40°C and 50°C, allowing crystal lattices to form in the paste.

Battery machine (BM) assembly and TBS:

- In the plate wrapping stage, positive plates are enveloped and interleaved with negative plates to form an element.
- The elements are held in the buffer area before they are transferred to the loading station, where the interleaved plate elements are loaded into the moulds.
- In the Cell Group Burning stage, positive and negative element lugs are welded together to form a cell.
- Cells are dropped into cases, and Through The Partition (TTP), welding takes place to weld each cell together.
- Covers/lids are heat-sealed onto the cases in the Heat-Sealing Station.
- In Post burning, the negative and positive end posts/terminals are automatically burnt before the batteries undergo a leak test. Liquid grease is applied to end post/terminals and the unfinished batteries are automatically packed into pellets.
- Extraction units serve various processes within BM assembly, which leads to stacks AX6, AX7, AX8 and AX22.
- Stack AX1 serves the TBS assembly area, while stack AX19 serves the La Pneumatic lid sealing machine.

Charge room (Battery formation and after charging):

- In the charge room, the batteries are filled with sulphuric acid. The Battery Acid Filling machine is served by four extraction units (AX20) that vent directly outside the building.
- The batteries are then charged as per applicable charging profiles.
- The batteries are washed and rinsed in washing tunnels before they are cleaned with chemicals and packaged for shipping.
- Stacks AX2 and AX3 serve the charge room.

Process emissions across the plant are captured at source by local extraction and filtration systems and vented to atmosphere through 40 stacks (including the addition of AX4) ranging from 4.0 m to 8.5 m in height. On this basis all process emissions are assumed to leave the plant via



the stacks and no fugitive emissions were modelled. Refer to Section 5.2 for details regarding emissions.

4 Dispersion modelling study

The Department of Environmental Affairs (DEA) homogenised dispersion modelling in South Africa by publishing “*Regulations Regarding Air Dispersion Modelling*” in Government Notice GN R.533 on 11 July 2014 (GN R.533). Throughout this report mention is made of compliance with this set of Regulations.

The dispersion modelling study was carried out with the US EPA AERMOD regulatory modelling system, which is listed as an approved dispersion model in GN R.533.

AERMOD is a steady-state Gaussian plume dispersion model that incorporates planetary boundary layer theory for both convective and stable atmospheric conditions. It is capable of simulating the dispersion of pollutants from point, area, volume and line sources over transport distances of up to approximately 50 km.

The following versions of the software were used: AERMOD v26135, AERMET v26135 and BPIP-PRIME v04274. AERMOD was applied in an urban dispersion environment, on a flat-terrain basis, with building downwash enabled.

In line with the per-calendar-year basis of the South African standards, AERMOD was run per meteorological year (2021 – 2025) and the multi-year result reported for each receptor is the maximum year (worst case) across the five years.

5 Model input

5.1 Meteorological data

AERMET requires a substantial quantity of meteorological data, including hourly surface data (wind speed, wind direction, temperature, humidity, pressure and cloud cover) and upper-air soundings. LAQS could not obtain a reliable set of surface measured data representative of the site and therefore used the ECMWF ERA5 reanalysis dataset, obtained via the Copernicus Climate Data Store, in place of a local weather station. Using simulated data is an acceptable method according to GN R.533.

Five consecutive calendar years (2021 – 2025) of hourly data were processed, satisfying the GN R.533 requirement of at least three consecutive years of data.

The distribution of winds, as determined from the ERA5 data, is shown in Figure 2.

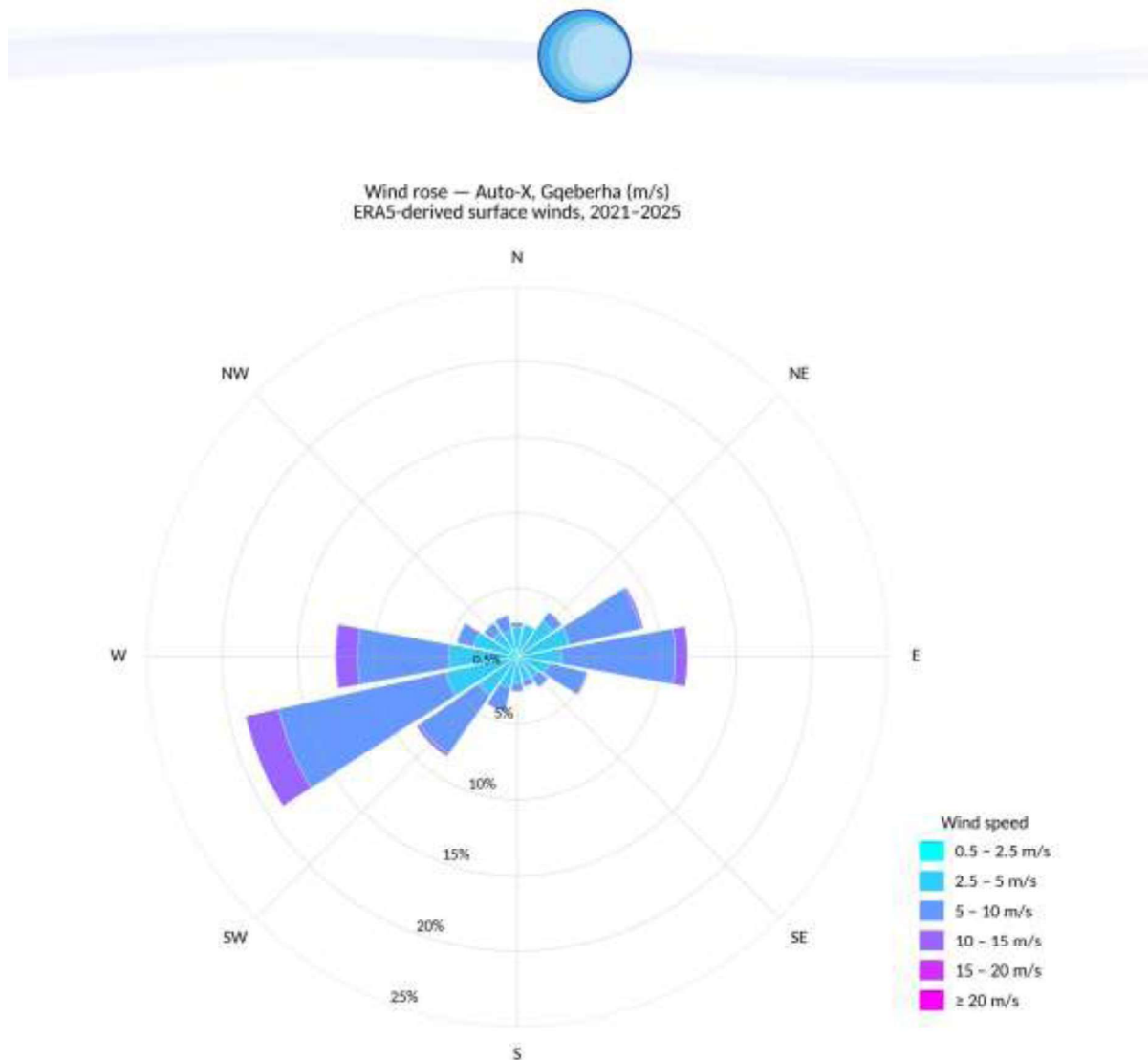


Figure 2: Distribution of wind directions, 2021 – 2025.

5.2 Emissions

As per GN R.533, for the purpose of an air quality impact assessment, it is assumed that emissions occur at the maximum permitted concentration as stated in the "*List of Activities That Result in Atmospheric Emissions*" as published in Government Notice 893 of 22 November 2013 (GN893). Thus, the emission rates are determined by the volumetric flow of each stack and the maximum permitted concentration of each regulated pollutant applicable to that stack.

Due to this difference in the emissions likely to occur, and the requirements of the regulations, LAQS modelled the following scenarios:

- **Scenario A:** Site only with emissions as per the AEL application form flow rates and the maximum permitted emission rates.
- **Scenario B:** Site only with emissions based on measured concentrations. For newly proposed stacks, emissions are estimated based on similar process measurements.
- **Scenario C:** Cumulative impact using site emissions as per Scenario B and actual emissions from nearby sources.



5.2.1 Particulate matter (PM) and lead (Pb)

Lead-bearing particulate matter is generated by lead oxide manufacturing, paste mixing, plate making, curing and the assembly operations. In the absence of supporting particle-size information, and in following a conservative approach, LAQS assumed that all particulates emitted conform to the definition of PM₁₀ particulate matter. This is an overestimation of PM₁₀ emissions as PM₁₀ forms a sub-set of total PM and is regarded as a worst-case scenario.

Lead is modelled as a separate pollutant.

5.2.2 Sulphur trioxide and sulphuric acid (H₂SO₄) mist

Sulphuric acid mist expressed as sulphur trioxide is a controlled pollutant potentially emitted from the formation charge rooms. The two compounds are related as sulphur trioxide rapidly forms sulphuric acid when it comes into contact with moisture.

Emissions in the form of sulphuric acid mist or sulphur trioxide (SO₃) may form during the battery charging processes. These emissions are captured by the extraction systems serving those areas and vented through the associated stacks.

For modelling purposes, all sulphur trioxides are assumed to be converted to sulphuric acid mist. This was not added to the modelled PM₁₀ concentrations.

Potential emissions from the handling and storage of sulphuric acid were considered in the inventory, however, these emissions were found to be negligible. The screening assessment conducted considered a series of highly conservative, worst-case scenarios across three primary emission pathways.

Evaporative losses were modelled using the Mackay & Matsugu (1973) formulation, assuming a total-loss-of-containment event where all containment bunds are permanently flooded to their full plan area with the strongest stored acids and exposed to ambient wind for a full year.

Vapour displacement during tank filling was calculated using US EPA AP-42 methodology (Section 5.2) assuming splash-loading for every litre transferred.

Finally, mechanical mist entrainment was constrained using a physical mass-balance bound, assuming the gas displaced during every transfer carried a maximum physically sustainable aerosol loading equivalent to a dense, opaque fog (1 000 mg/m³).

Under these extreme bounding conditions, total area-source emissions were found to be negligible, amounting to approximately 19 kg of sulphuric acid per year (15.56 kg SO₃ equivalent). Sodium hydroxide was excluded from the evaporative calculations due to its non-volatile nature. The combined evaporative and mechanical emissions from the storage area represent just 0.05% of the facility's licensed point-source SO₃ emissions and 0.003% of total SO_x. Consequently, the assessment demonstrates that routine and evaporative emissions from the storage tanks are immaterial to the site's overall environmental footprint and do not warrant inclusion in the formal atmospheric dispersion model.



5.2.3 Sulphur dioxide and nitrogen dioxide

Sulphur dioxide (SO₂) and nitrogen dioxide (NO₂) emissions are reported in the emissions inventory for the charge room extraction stacks, the battery acid filling stacks and the proposed expansion oven stack.

Emissions were derived from the minimum emission standards applicable to the plant rather than from measured concentrations. Where a stack is subject to both an SO₂ and a NO_x standard at the same concentration limit, the derived mass emission rates for the two pollutants are consequently identical. This is a direct result of the limit-based derivation and is the intended conservative treatment; it does not indicate duplicated data.

It is however noted that SO₂ and NO₂ emissions are not expected from the formation charging process. Their formation requires either the reduction of strong chemical bonds or thermal decomposition at high temperature, and neither occurs in a charge room.

Formation charging is an electrochemical process: it evolves hydrogen and oxygen at the plates and carries an entrained sulphuric acid electrolyte mist out of the cells with the evolved gas. It involves no combustion and produces neither oxide.

The 2025 stack survey supports the commentary: SO₂ was not detected in AX2 and only trace amounts were detected in AX3. No NO₂ was detected in either stack.

This is significant, as 84% of the modelled SO₂ and NO_x originate from these two stacks.

The emissions were nonetheless modelled as listed. The Regulations require the assessment to be carried out at the maximum permitted concentration for the pollutants the licence lists against each stack, and it is not open to the assessment to reallocate them. The consequence is recorded here so that the predicted SO₂ and NO₂ concentrations are read as a property of the licence schedule rather than as a prediction of the plant's behaviour.

In the absence of measured air quality data, this assessment applies the Tier 2 Ambient Ratio Method with the national NO₂/NO_x ratio of 0.80 prescribed by the regulations. Predicted NO₂ concentrations are therefore 80% of the modelled NO_x concentrations at every receptor and averaging period.

5.3 Emission rates

Emissions were modelled as continuous (24 hours per day, 365 days per year), which overstates long-term averages relative to the actual operating cycle and is therefore conservative. The rates for each scenario were derived as follows:

Scenario A: Site only emissions at maximum permitted emission rates

The plant was modelled as 40-point sources listed on the AEL. Total emission rates by pollutant are determined based on the volumetric flow rate of each stack listed in the AEL application and the maximum permitted controlled pollutant concentration in each of these point sources. The combined emission rates are summarised in Table 3; the full stack parameter and emission rate schedule is provided in Appendix A.



Scenario B: Site only emissions – actual emission rates

The existing AutoX stacks are modelled at the rates and stack conditions recorded in the 2025 stack-testing campaign, and the proposed expansion stacks emissions are estimated from similar units at the plant. It differs from Scenario A as it represents the emissions likely to occur from the plant instead of the maximum licensed emissions.

Scenario C: Cumulative emissions

The existing AutoX stacks at the emission rates as measured and estimated in Scenario B along with actual emissions from industrial sources in the vicinity of the plant.

Pollutant	Scenario A total (g/s)	Scenario B total (g/s) / AutoX share	Scenario C total (g/s) / AutoX share
PM ₁₀	2.52	0.194 (100%)	0.594 (33%)
Pb	0.17	0.0201 (100%)	0.0201 (100%)
SO ₂	13.55	0.0189 (100%)	2.47 (0.8%)
NO ₂	13.55	0.00492 (100%)	0.409 (1.2%)
H ₂ SO ₄	1.18	0.00863 (100%)	0.00863 (100%)
Point sources modelled	40	40 (40 AutoX)	50 (40 AutoX)

Table 3: Modelled emission rates by pollutant and scenario. The percentage is the AutoX share of the cumulative total.

The annual masses implied by the Scenario A rates, at continuous operation, are:

PM ₁₀	79.8 tons
Pb	5.3 tons
SO ₂	427 tons
NO ₂	427 tons
H ₂ SO ₄	37.4 tons

The annual masses implied by the Scenario B rates, at continuous operation, are:

PM ₁₀	6.1 tons
Pb	0.6 tons
SO ₂	0.6 tons
NO ₂	0.2 tons
H ₂ SO ₄	0.3 tons



Refer to Appendix A for details regarding the sources included.

The surrounding sources include 10 permitted stacks drawn from the LAQS emissions database of surrounding operations. They are modelled at their own measured emission rates and stack conditions.

The annual masses contributed by other sources considered in the cumulative assessment (Scenario C), based on actual emission at continuous operation are:

PM ₁₀	12.6 tons
Pb	0.09 tons
SO ₂	77.4 tons
NO ₂	12.7 tons

5.4 Receptor grid and model domain

A multi-tier Cartesian receptor grid centred on the site was used, in accordance with GN R.533: 50 m spacing out to 1 000 m from site centre and 100 m spacing out to 5 000 m. Receptors were placed at ground level (flagpole height 0.0 m).

6 Compliance framework

National Ambient Air Quality Standards (NAAQS) are published in Government Notice No. 1210 of 24 December 2009 (GN1210) and subsequent amendments. Predicted concentrations are assessed against the NAAQS for PM₁₀, Pb, SO₂ and NO₂. All particulate emissions were conservatively treated as PM₁₀ (Section 5.2.1).

The NAAQS do not include sulphuric acid. Predicted H₂SO₄ concentrations are therefore compared against the Reference Exposure Levels (RELs) published by the Office of Environmental Health Hazard Assessment of the California Environmental Protection Agency (OEHHA), which are health-based ambient reference values derived for the protection of the general population. OEHHA publishes an acute REL of 120 µg/m³ (1-hour averaging period) and a chronic REL of 1 µg/m³ (annual average) for sulphuric acid, both based on respiratory endpoints.

It is recommended that the 99th percentile concentration is used for assessing compliance with short-term NAAQS, on the basis that the single highest predicted ground-level concentration can be an outlier arising from the variability of the modelled meteorology. The 99th percentile is extracted directly by AERMOD as a ranked value: rank 4 for a 24-hour averaging period (365 values per year) and rank 88 for a 1-hour period (8 760 values per year). Annual averages are assessed against the highest annual mean, with no exceedances permitted.

The assessment criteria are summarised in Table 4.



Pollutant	Averaging period	Limit ($\mu\text{g}/\text{m}^3$)	Permitted frequency of exceedance	Compliance statistic modelled
PM ₁₀	24 hours	75	4 per year	99th percentile daily value (rank 4), worst year
PM ₁₀	1 year	40	0	Highest annual mean, worst year
Pb	1 year	0.5	0	Highest annual mean, worst year
SO ₂	1 hour	350	88 per year	99th percentile hourly value (rank 88), worst year
SO ₂	24 hours	125	4 per year	99th percentile daily value (rank 4), worst year
SO ₂	1 year	50	0	Highest annual mean, worst year
NO ₂	1 hour	200	88 per year	99th percentile hourly value (rank 88), worst year
NO ₂	1 year	40	0	Highest annual mean, worst year
H ₂ SO ₄ (SO ₃)	1 hour	120 (OEHHA acute REL, screening)	—	Maximum hourly value (rank 1), worst year
H ₂ SO ₄ (SO ₃)	1 year	1 (OEHHA chronic REL, screening)	—	Highest annual mean, worst year

Table 4: Assessment criteria.

7 Results

The three scenarios modelled are summarised below in Table 5, as discussed in Section 5.

Scenario	Basis	Relevant table and figures
Scenario A	Site only, with emissions derived from the flow rates given in the AEL application form and the maximum permitted emission concentration applicable to each stack.	Tables 6 and 7; Figures 3 to 12
Scenario B	Site only, using the emissions the plant is expected to release — the existing stacks at their 2025 measured rates and the proposed expansion stacks at rates estimated from measured analogues on the same plant.	Table 8; Figures 13 to 22
Scenario C	Cumulative impact, using the Scenario B site emissions together with the actual emissions of the surrounding sources.	Table 9; Figures 23 to 32

Table 5: Emission scenarios assessed

Note: The legend scales are not common for all three scenarios. The base levels for Scenario B and C are adjusted to accommodate the lower emission rates.



7.1 Scenario A — site only, AEL application basis

Site only, with emissions derived from the flow rates given in the AEL application form and the maximum permitted emission concentration applicable to each stack.

Pollutant / statistic	Limit (µg/m³)	Maximum (µg/m³)	Location of maximum from site centre	Maximum as % of limit	Maximum at a sensitive receptor (µg/m³)
PM ₁₀ , 24-hour 99th percentile	75	168	220 m ESE	224%	41 (54%)
PM ₁₀ , annual average	40	77	160 m ESE	192%	6.9 (17%)
Pb, annual average	0.5	5.10	160 m ESE	1020%	0.46 (92%)
SO ₂ , 1-hour 99th percentile	350	1 753	180 m ENE	501%	325 (93%)
SO ₂ , 24-hour 99th percentile	125	792	180 m ENE	634%	170 (136%)
SO ₂ , annual average	50	359	50 m ESE	719%	23 (46%)
NO ₂ , 1-hour 99th percentile	200	1 196	50 m ENE	598%	148 (74%)
NO ₂ , annual average	40	288	50 m ESE	719%	19 (47%)
H ₂ SO ₄ , maximum 1-hour average	120 (*)	274	50 m ENE	228%	58 (48%)
H ₂ SO ₄ , annual average	1 (*)	31	50 m ESE	3 110%	1.99 (199%)

(*): OEHHA acute REL, screening

Table 6: Scenario A — predicted ground-level concentrations (worst year, 2021 – 2025). Values above the applicable criterion are shown in red.



Sensitive receptor	PM ₁₀ (24-h)	PM ₁₀ (annual)	Pb (annual)	SO ₂ (1-h)	SO ₂ (24-h)	SO ₂ (annual)	NO ₂ (1-h)	NO ₂ (annual)	H ₂ SO ₄ (1-h)	H ₂ SO ₄ (annual)
Livingstone Hospital Primary School	38	5.7	0.38	322	170	22	105	17	54	1.85
Livingstone Hospital	36	5.9	0.39	325	170	23	130	19	51	1.99
Frank Joubert Primary School	41	6.9	0.46	311	133	19	99	15	58	1.61
De Vos Malan Primary School	22	5.0	0.33	222	105	20	148	16	37	1.74
Life Mercantile Private Hospital	24	4.4	0.29	172	70	15	123	12	34	1.30
G J Louw Primary School	13	1.8	0.12	124	54	6.9	39	5.7	25	0.61
Adolph Schauder Primary School	13	2.6	0.17	123	55	12	75	10	21	1.09
Abraham Levy Primary School	13	2.6	0.17	123	55	12	75	10	21	1.09
Phakamisa High School	9.1	1.0	0.06	70	35	3.8	5.4	3.1	22	0.34
St Teresa's (RC) Primary School	12	1.1	0.07	82	46	4.3	6.6	3.5	20	0.38
St James (RC) Secondary School	11	1.1	0.07	86	40	4.2	5.5	3.5	21	0.37



Sensitive receptor	PM ₁₀ (24-h)	PM ₁₀ (annual)	Pb (annual)	SO ₂ (1-h)	SO ₂ (24-h)	SO ₂ (annual)	NO ₂ (1-h)	NO ₂ (annual)	H ₂ SO ₄ (1-h)	H ₂ SO ₄ (annual)
Dietrich Primary School	8.8	1.5	0.10	97	41	6.6	49	5.4	21	0.59
Empilweni TB Hospital	8.49	1.13	0.08	75	32	4.79	36	3.96	19	0.43
Sydenham Primary School	9.42	1.25	0.08	89	33	4.82	47	4.01	17	0.43
Ntyatyambo Primary School	4.82	0.54	0.04	40	19	2.21	3.21	1.86	15	0.20
Emzomncane Primary School	5.14	0.53	0.04	42	21	2.24	2.92	1.90	14	0.21
Criterion (µg/m³)	75	40	0.5	350	125	50	200	40	120	1

Table 7: Scenario A — predicted concentrations at each sensitive receptor (µg/m³, worst year 2021 – 2025). Values above the applicable criterion are shown in red.

PM10 — 24-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

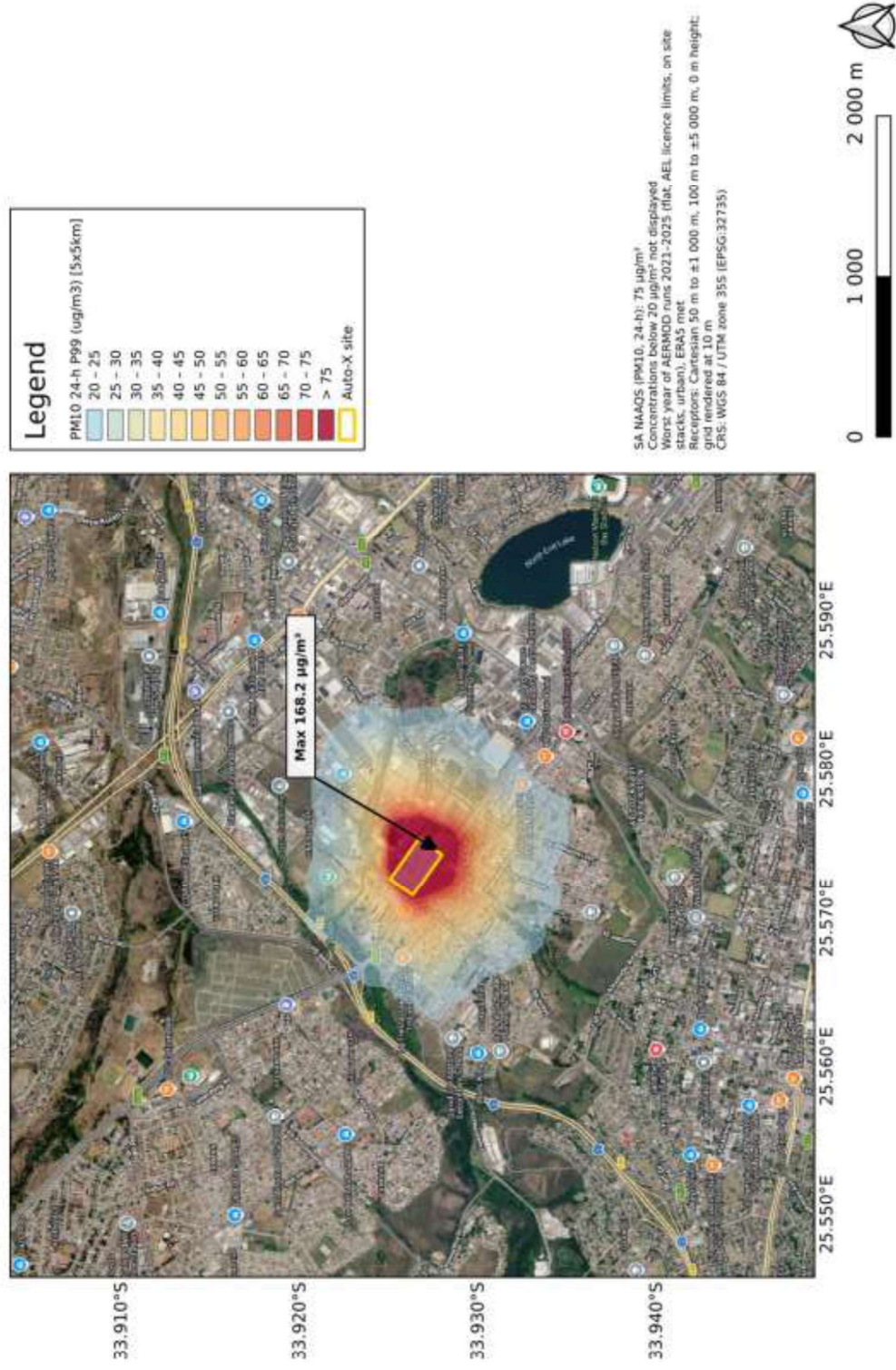


Figure 3: Scenario A — 99-percentile daily averaged PM₁₀ concentrations, µg/m³. Legend scaled to the NAAQS of 75 µg/m³ in classes of 5 µg/m³; concentrations below 20 µg/m³ not displayed; map extent 5 km x 5 km.

PM10 — Annual average

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

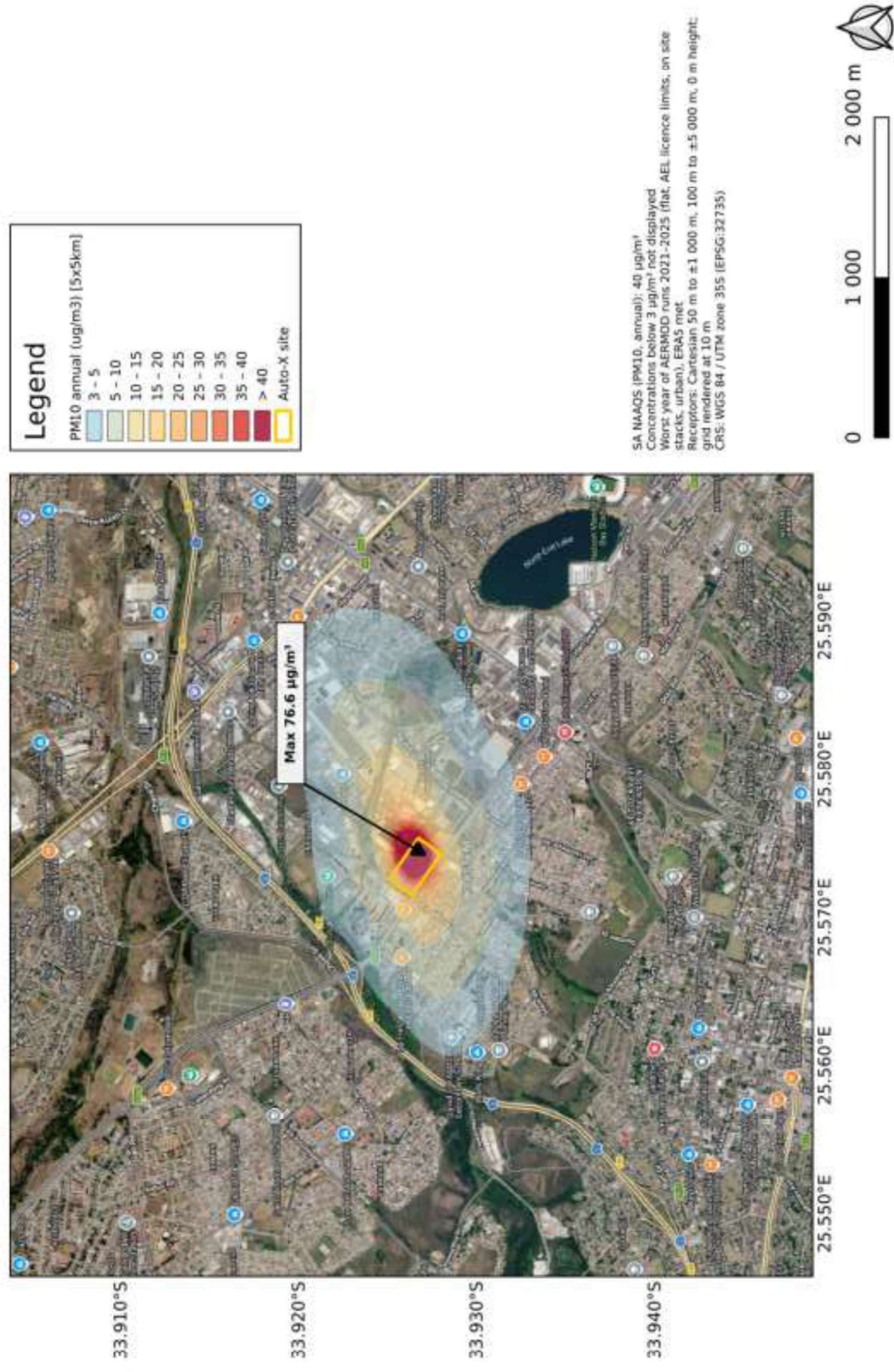


Figure 4: Scenario A — annual averaged PM₁₀ concentrations, µg/m³. Legend scaled to the NAAQS of 40 µg/m³ in classes of 5 µg/m³; concentrations below 3 µg/m³ not displayed; map extent 5 km x 5 km.

Pb — Annual average

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

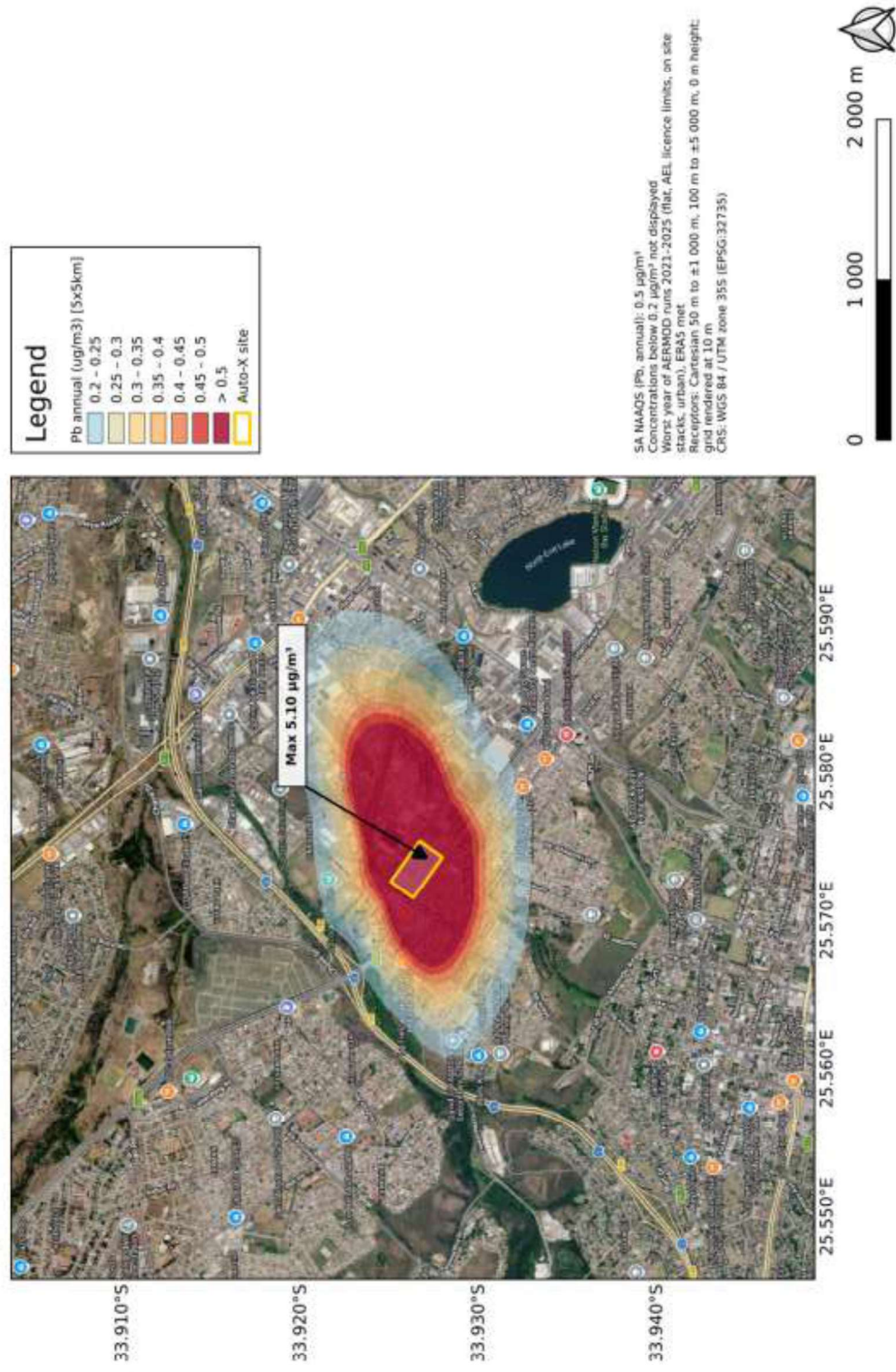


Figure 5: Scenario A — annual averaged Pb concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the NAAQS of $0.5 \mu\text{g}/\text{m}^3$ in classes of $0.05 \mu\text{g}/\text{m}^3$; concentrations below $0.2 \mu\text{g}/\text{m}^3$ not displayed; map extent 5 km x 5 km.

SO2 — 1-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

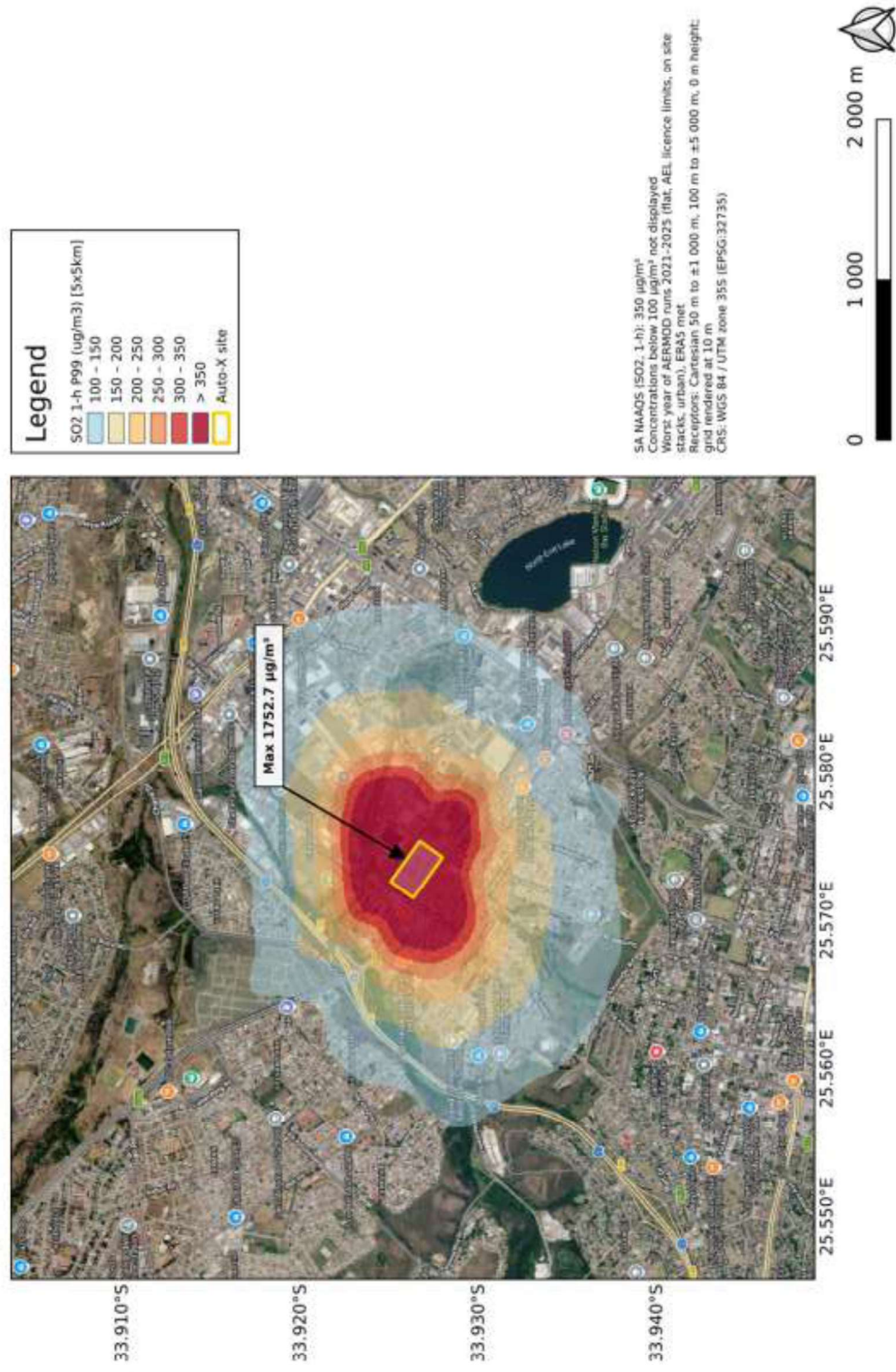


Figure 6: Scenario A — 99-percentile hourly SO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 350 µg/m³ in classes of 50 µg/m³; concentrations below 100 µg/m³ not displayed; map extent 5 km x 5 km.

SO2 — 24-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

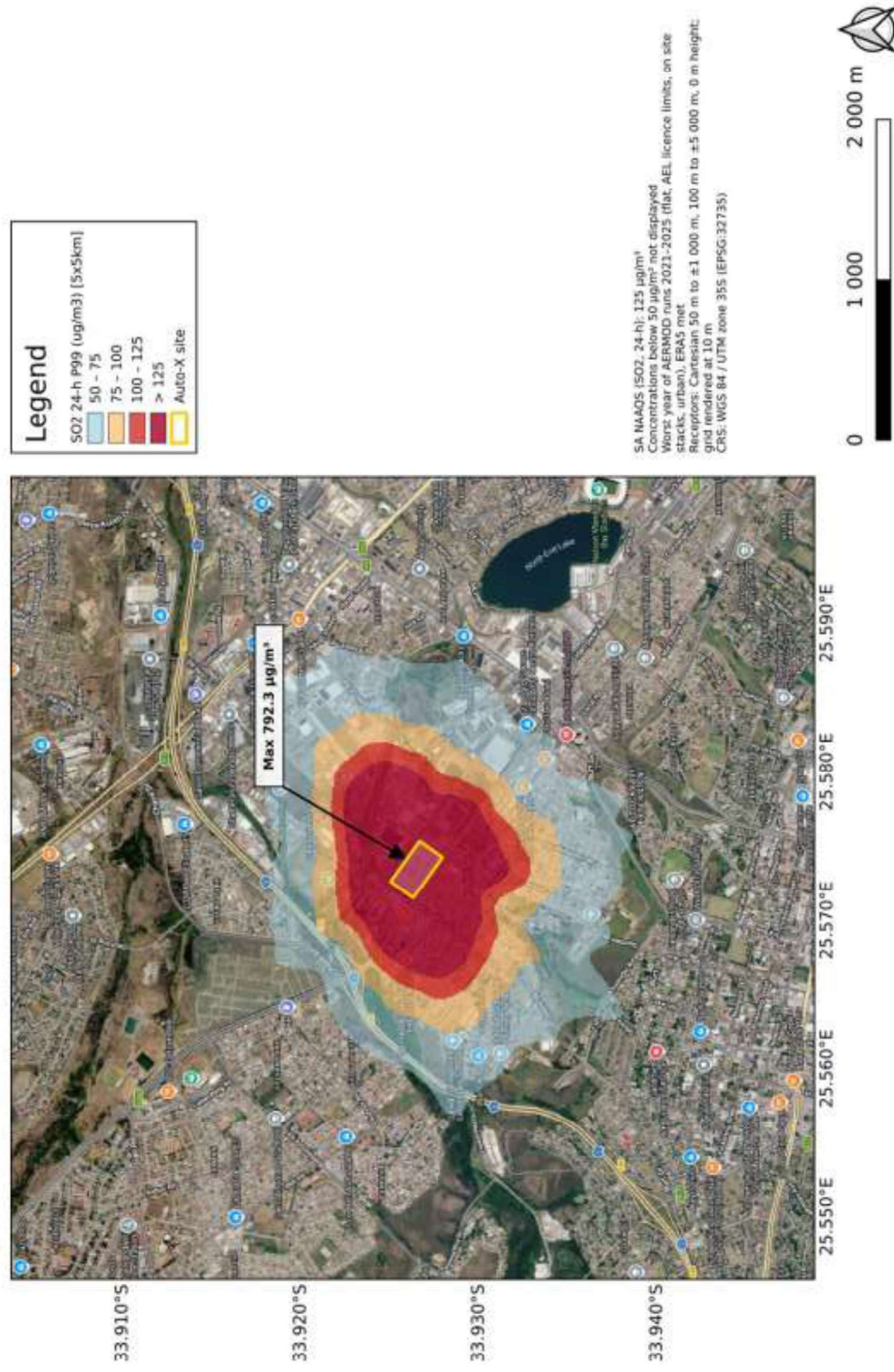


Figure 7: Scenario A — 99-percentile daily averaged SO₂ concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the NAAQS of 125 $\mu\text{g}/\text{m}^3$ in classes of 25 $\mu\text{g}/\text{m}^3$; concentrations below 50 $\mu\text{g}/\text{m}^3$ not displayed; map extent 5 km x 5 km.

SO2 — Annual average

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

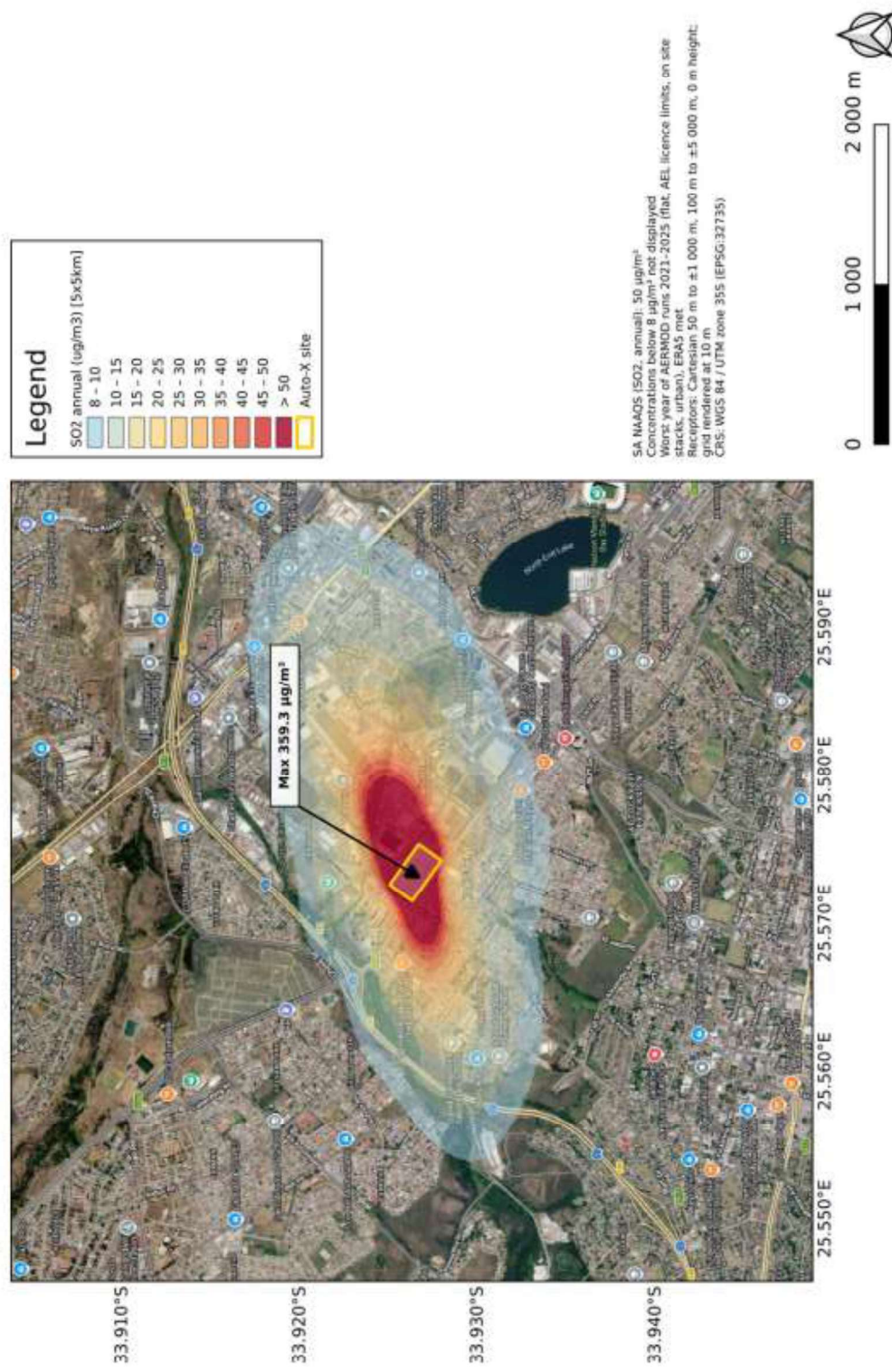


Figure 8: Scenario A — annual averaged SO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 50 µg/m³ in classes of 5 µg/m³; concentrations below 8 µg/m³ not displayed; map extent 5 km x 5 km.

NO2 — 1-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

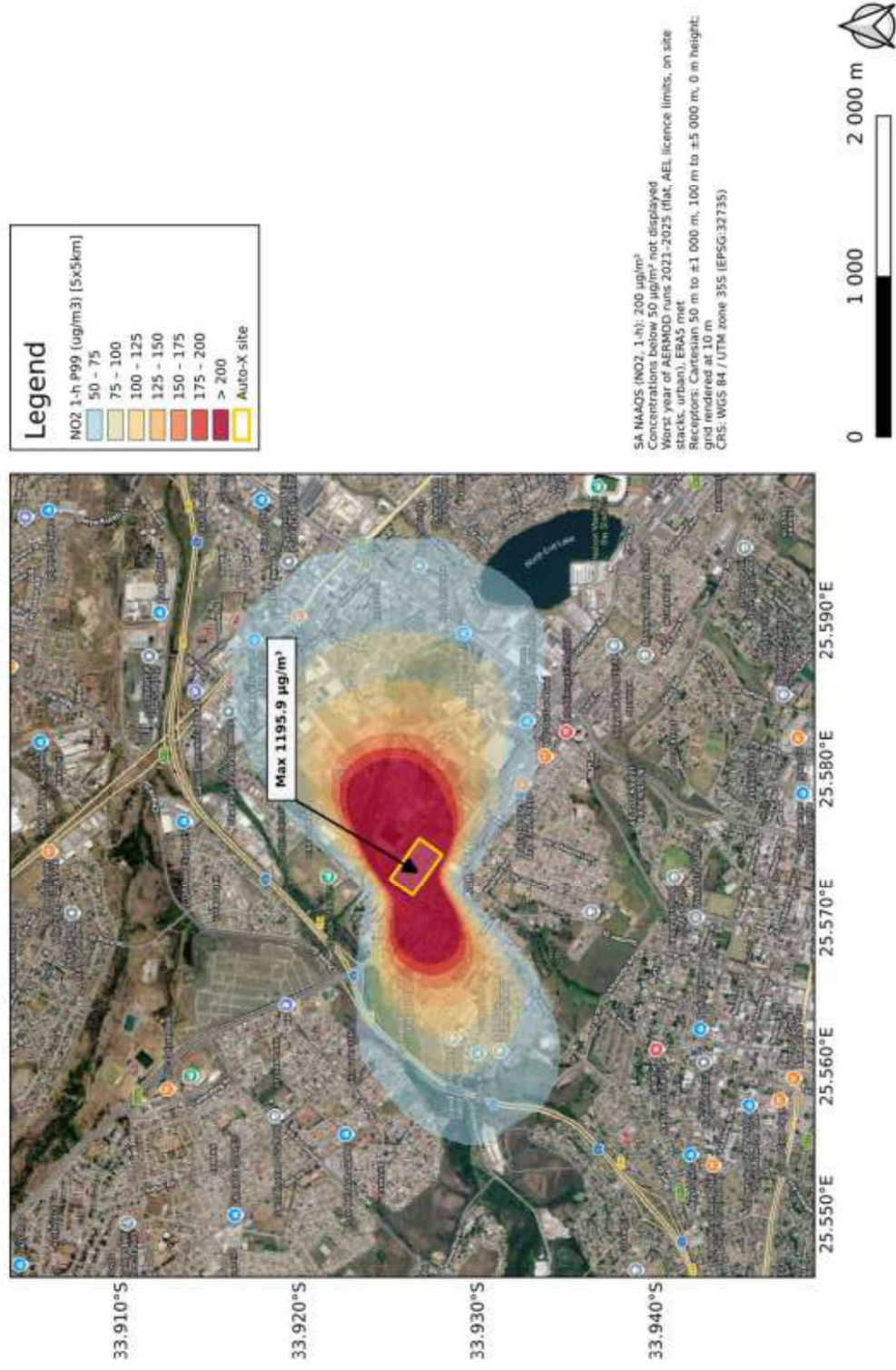


Figure 9: Scenario A — 99-percentile hourly NO₂ concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the NAAQS of 200 $\mu\text{g}/\text{m}^3$ in classes of 25 $\mu\text{g}/\text{m}^3$; concentrations below 50 $\mu\text{g}/\text{m}^3$ not displayed; map extent 5 km x 5 km.

NO2 — Annual average

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

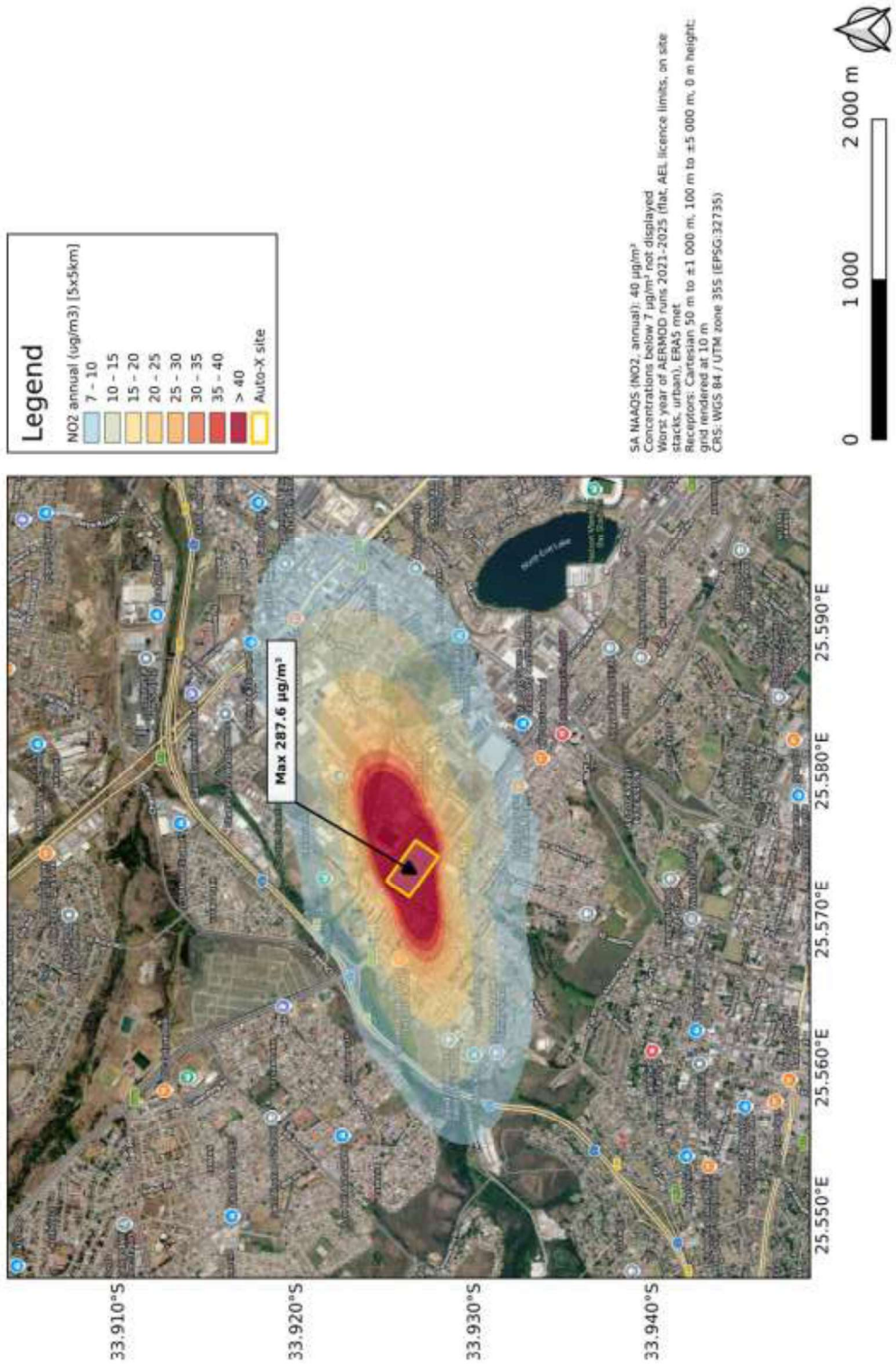


Figure 10: Scenario A — annual averaged NO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 40 µg/m³ in classes of 5 µg/m³; concentrations below 7 µg/m³ not displayed; map extent 5 km x 5 km.

H2SO4 — Annual average

Auto-X, Gqeberha — Model basis: AEL (flat, AEL licence limits, on site stacks)

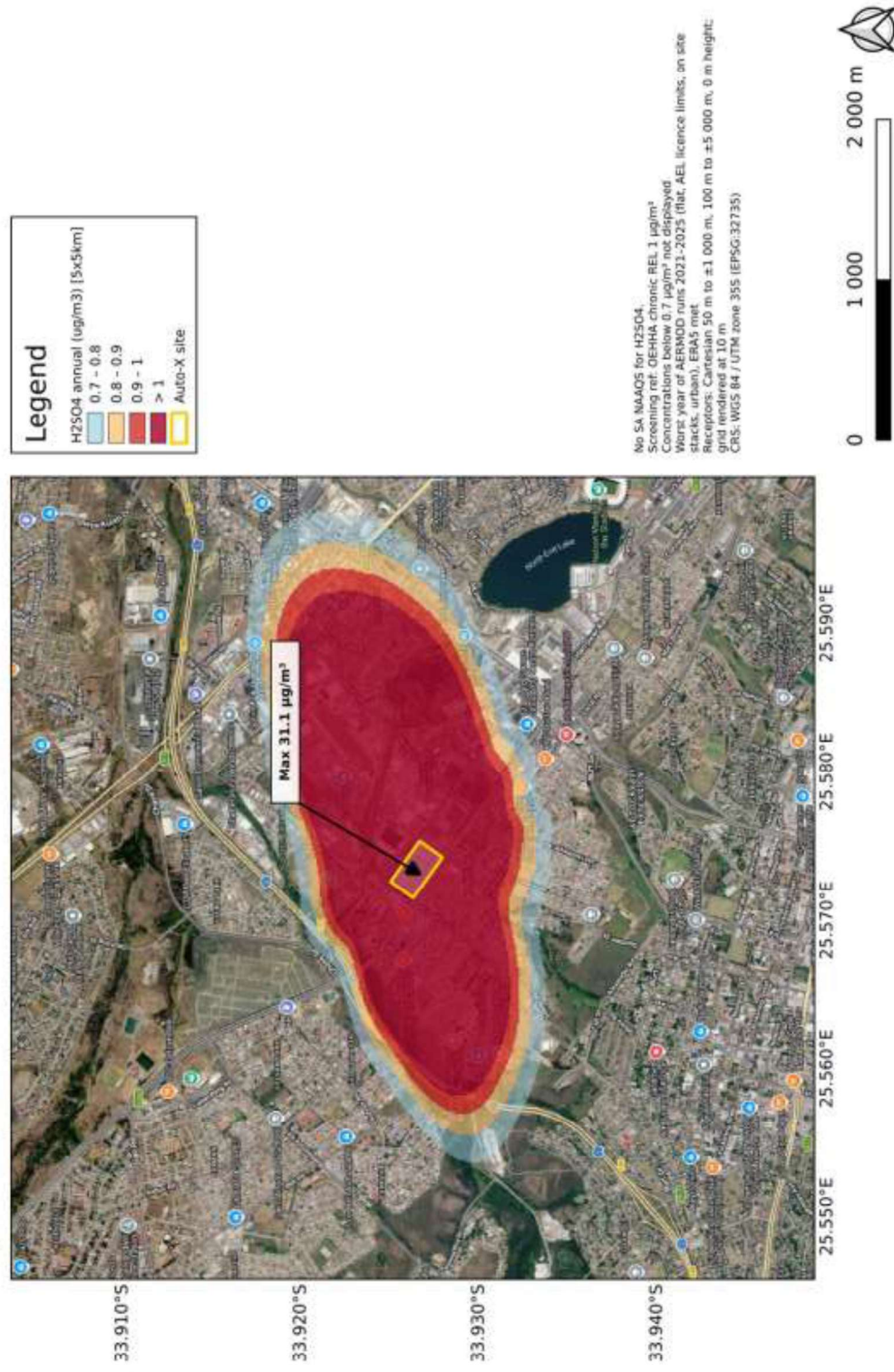


Figure 12: Scenario A — annual averaged H₂SO₄ concentrations, µg/m³. Legend scaled to the OEHHA chronic REL of 1 µg/m³ in classes of 0.1 µg/m³; concentrations below 0.7 µg/m³ not displayed; map extent 5 km x 5 km.



7.2 Scenario B — site only, actual expected emissions

Scenario B refers to the actual emissions emitted by the site, as measured in 2025 and includes the new proposed stacks with their emissions estimated from similar emissions units at the plant. As can be seen from Table 8, no significant impact is expected at any of the sensitive receptors. For lead, the only pollutant exceeding the NAAQS, the highest concentration at a sensitive receptor is 13% of the NAAQS. This occurs at the Frank Joubert Primary School with all other receptors measuring lower.

Pollutant / statistic	Limit (µg/m³)	Maximum (µg/m³)	Location of maximum from site centre	Maximum as % of limit	Maximum at a sensitive receptor (µg/m³)
PM ₁₀ , 24-hour 99th percentile	75	12.2	240 m SE	16%	2.8 (4%)
PM ₁₀ , annual average	40	5.2	150 m E	13%	0.5 (1%)
Pb, annual average	0.5	1.05	250 m ESE	211%	0.06 (13%)
SO ₂ , 1-hour 99th percentile	350	7.6	15 m SE	2%	1.11 (0.3%)
SO ₂ , 24-hour 99th percentile	125	4.2	15 m SE	3%	0.51 (0.4%)
SO ₂ , annual average	50	1.1	14 m SE	2%	0.07 (0.1%)
NO ₂ , 1-hour 99th percentile	200	1.4	14 m SE	0.7%	0.09 (<0.1%)
NO ₂ , annual average	40	0.2	14 m SE	0.6%	0.02 (<0.1%)
H ₂ SO ₄ , maximum 1-hour average	120 (OEHHA acute REL, screening)	3.4	50 m ESE	3%	0.61 (0.5%)
H ₂ SO ₄ , annual average	1 (OEHHA chronic REL, screening)	0.21	230 m ENE	21%	0.01 (1%)

Table 8: Scenario B — predicted ground-level concentrations (worst year, 2021 – 2025).
Values above the applicable criterion are shown in red

PM10 — 24-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

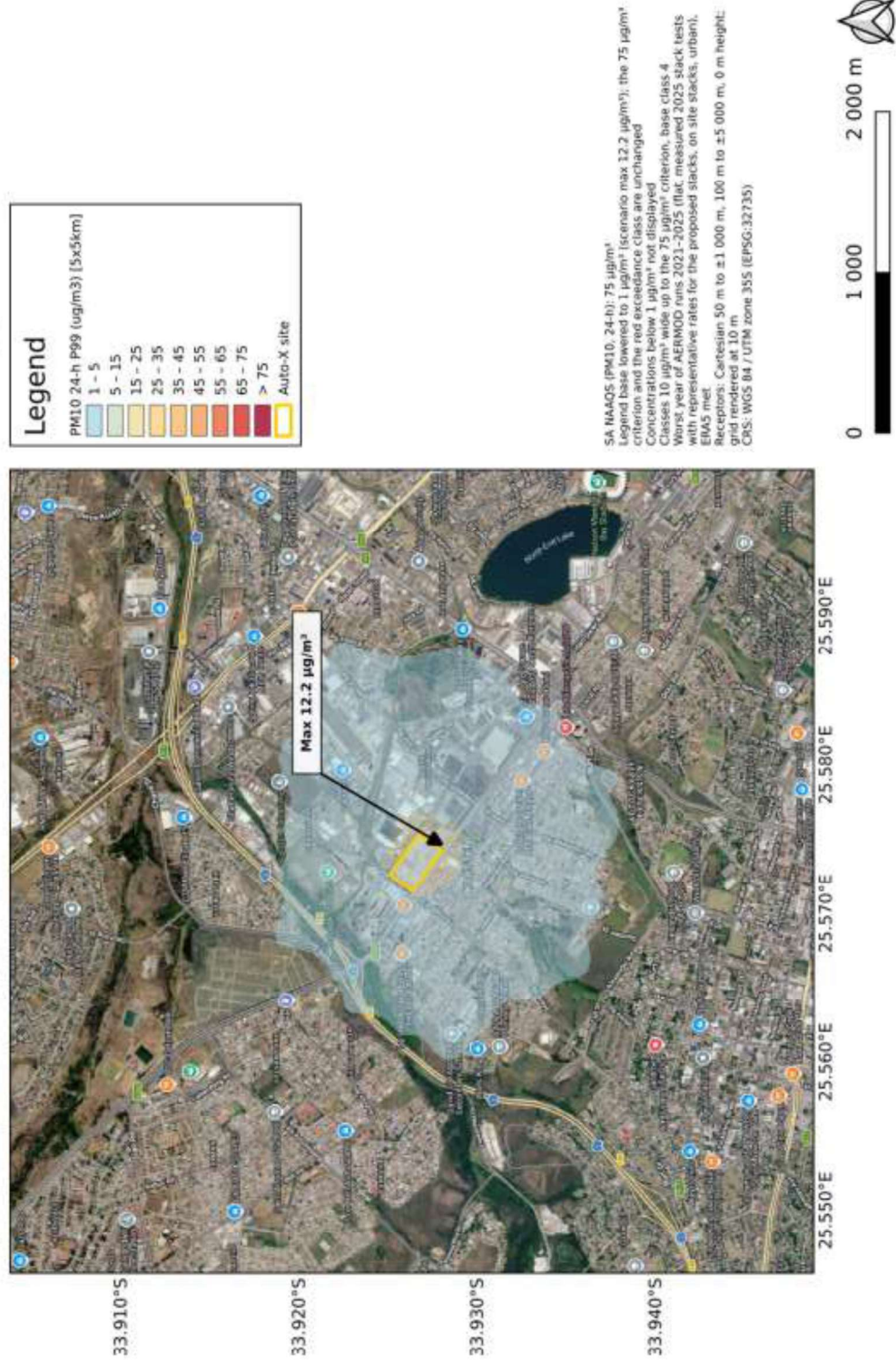


Figure 13: Scenario B — 99-percentile daily averaged PM₁₀ concentrations, µg/m³. Legend scaled to the NAAQS of 75 µg/m³ in classes of 5 µg/m³; concentrations below 1 µg/m³ not displayed; map extent 5 km x 5 km.

PM10 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

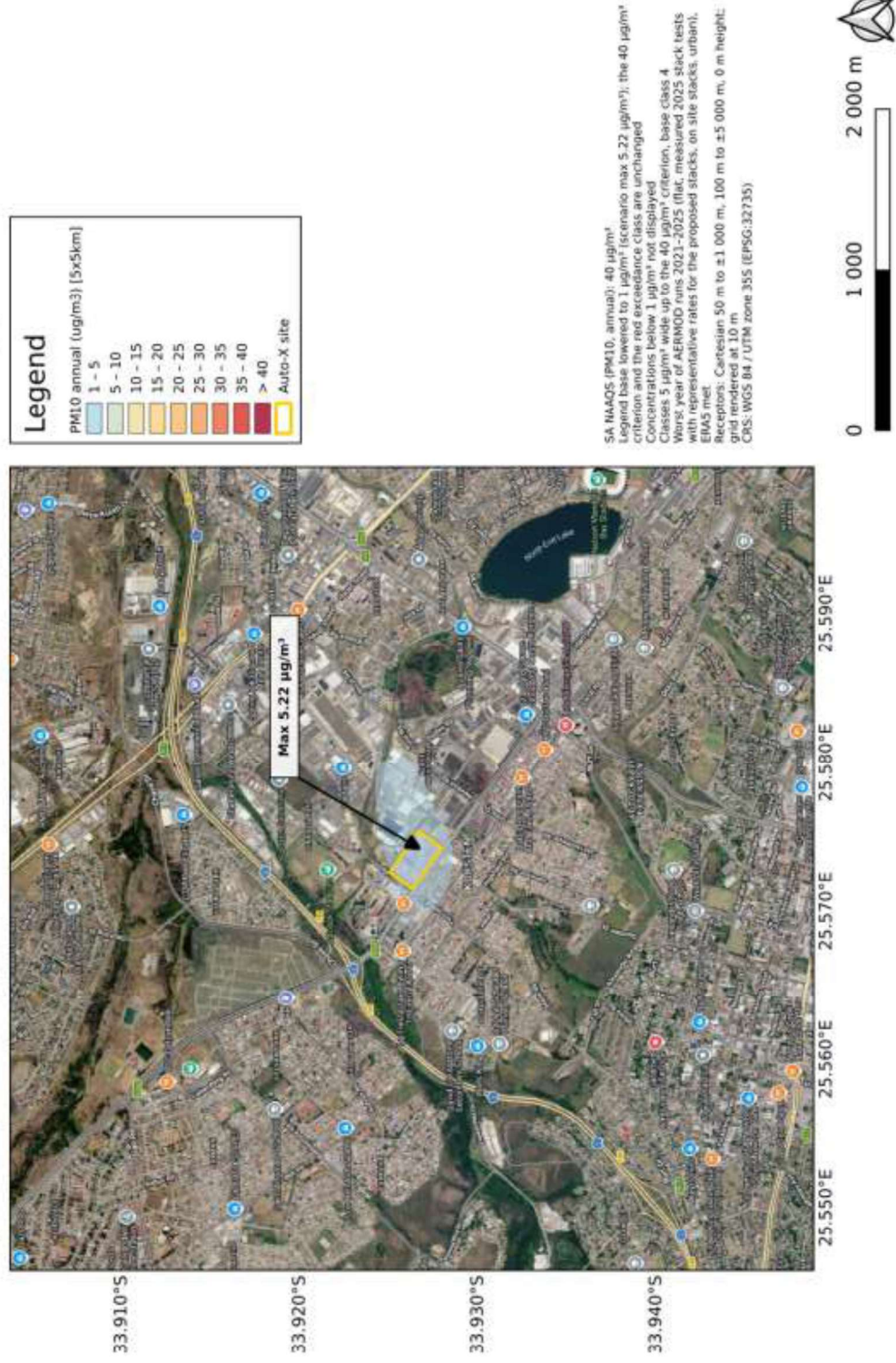


Figure 14: Scenario B — annual averaged PM₁₀ concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the NAAQS of 40 $\mu\text{g}/\text{m}^3$ in classes of 5 $\mu\text{g}/\text{m}^3$; concentrations below 1 $\mu\text{g}/\text{m}^3$ not displayed; map extent 5 km x 5 km.

Pb — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

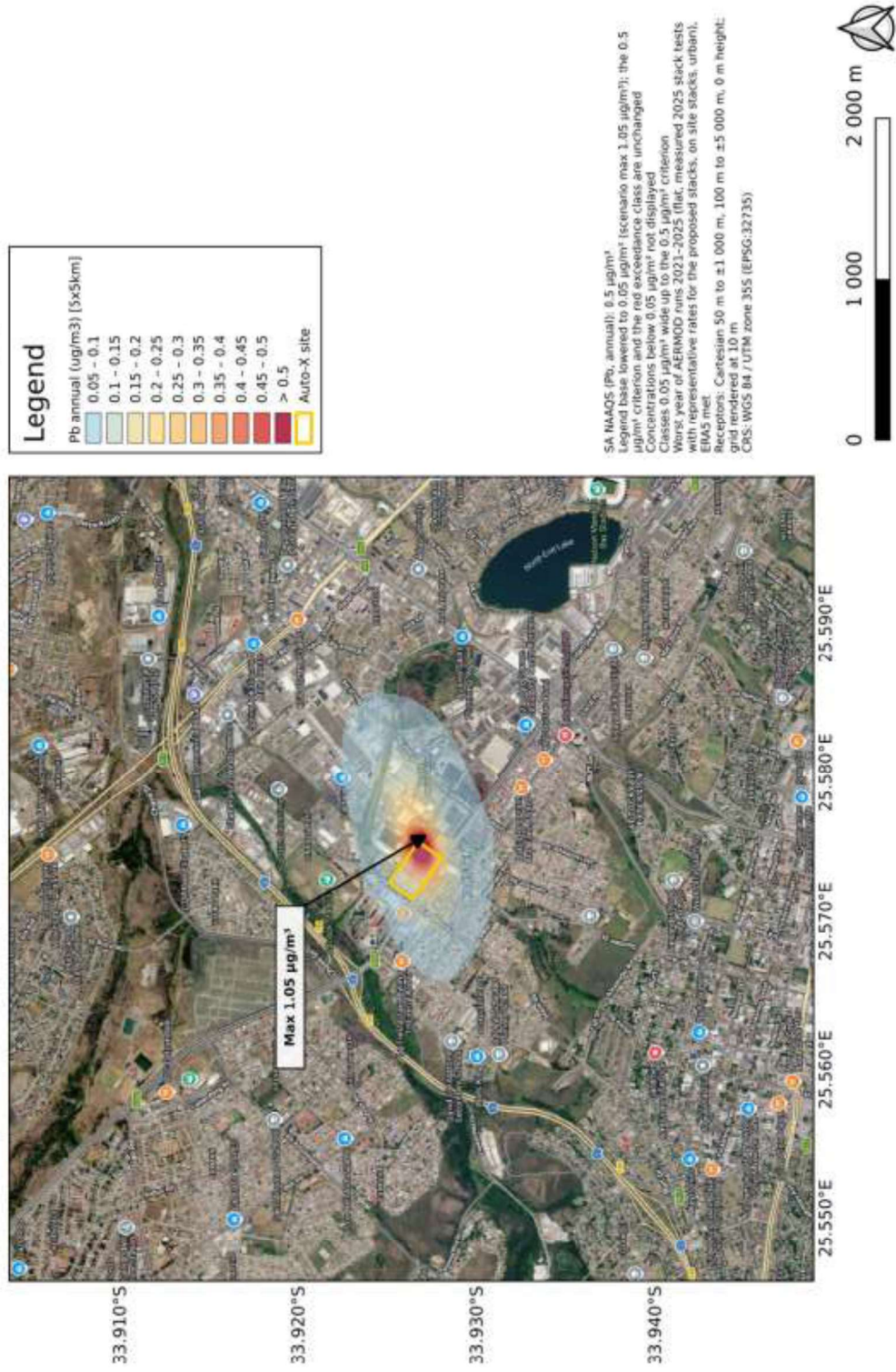
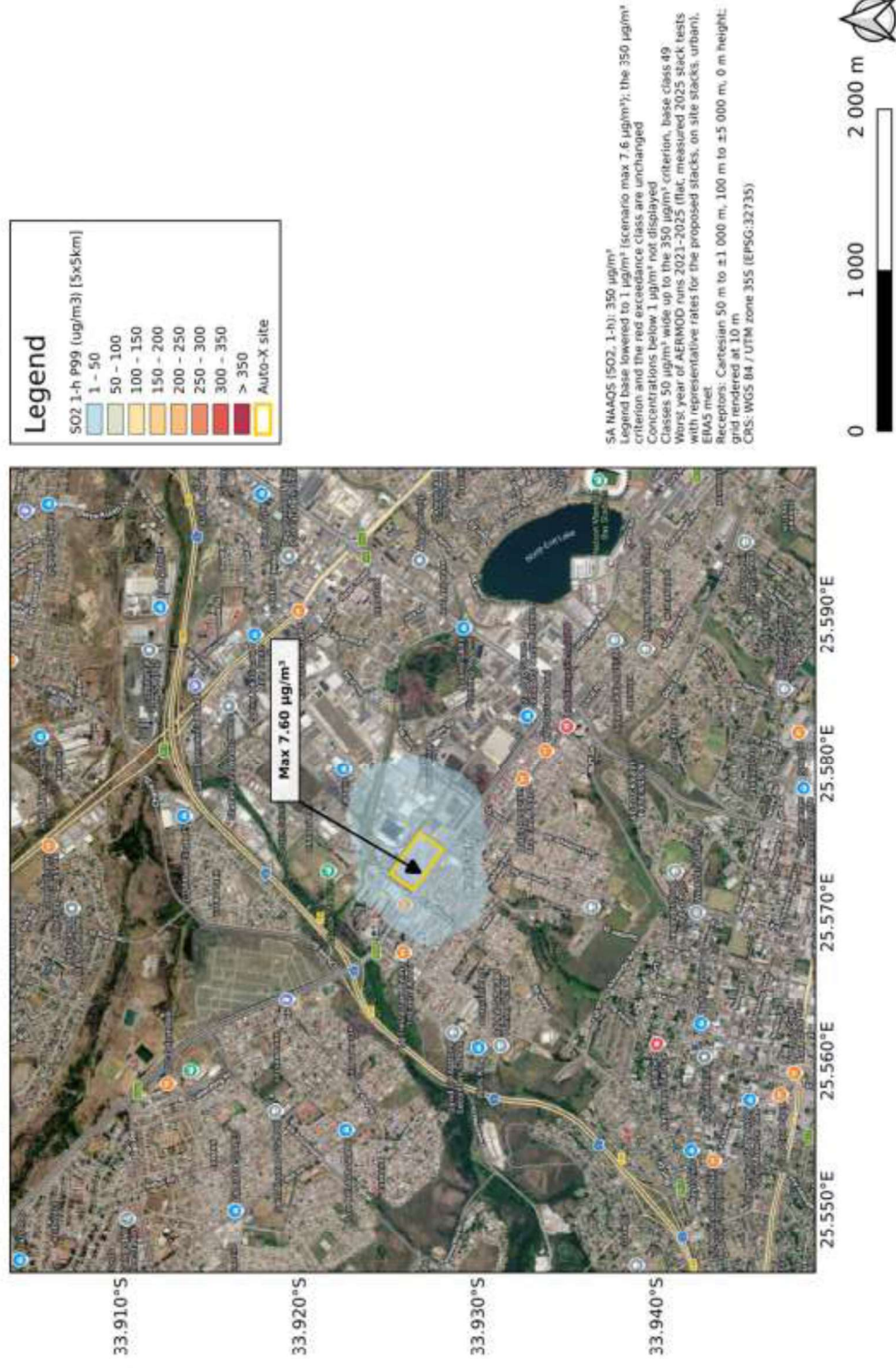


Figure 15: Scenario B — annual averaged Pb concentrations, µg/m³. Legend scaled to the NAAQS of 0.5 µg/m³ in classes of 0.05 µg/m³; concentrations below 0.05 µg/m³ not displayed; map extent 5 km x 5 km.

SO2 — 1-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)



SO2 — 24-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

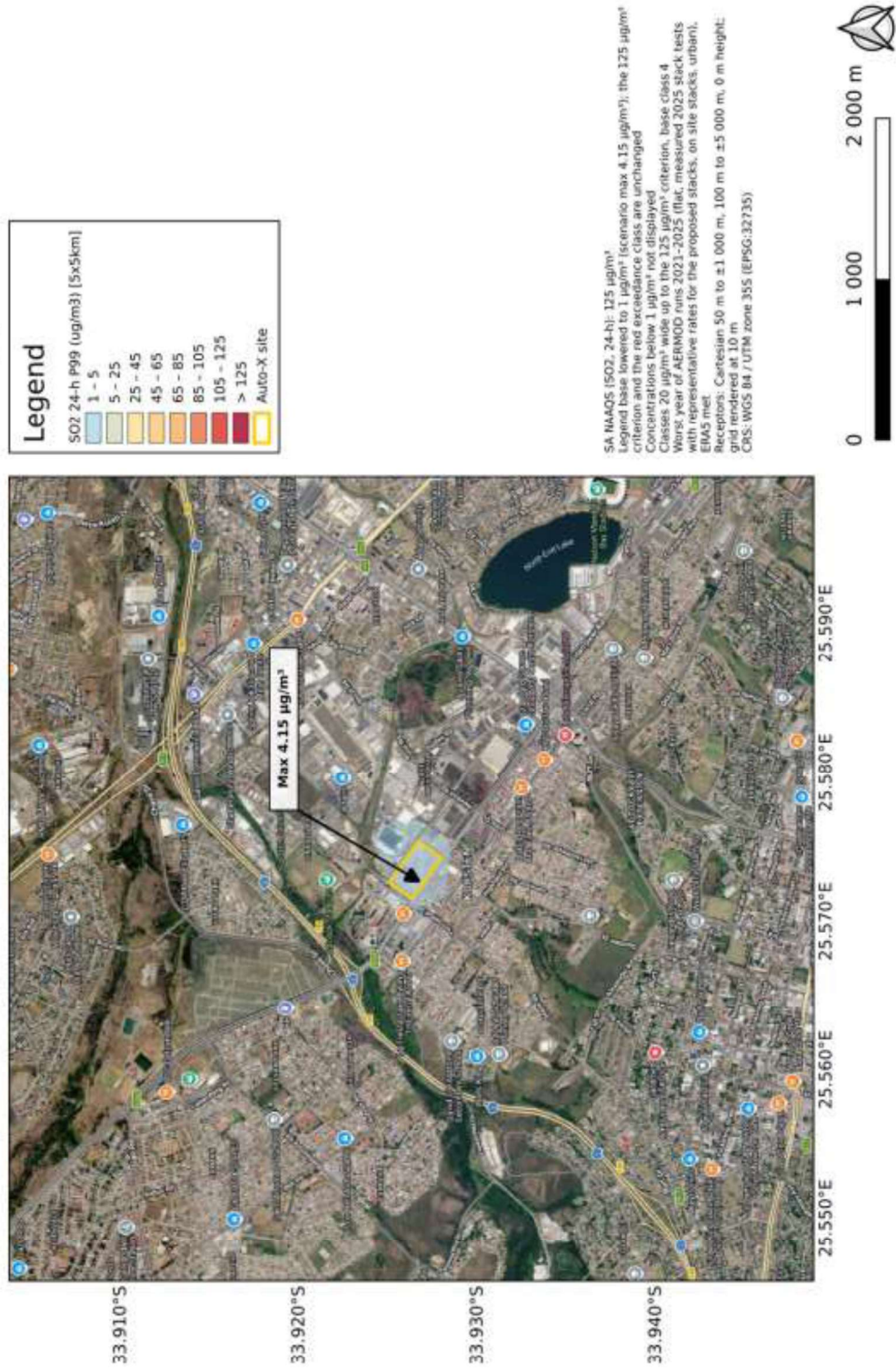


Figure 17: Scenario B — 99-percentile daily averaged SO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 125 µg/m³ in classes of 20 µg/m³; concentrations below 1 µg/m³ not displayed; map extent 5 km x 5 km.

SO2 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

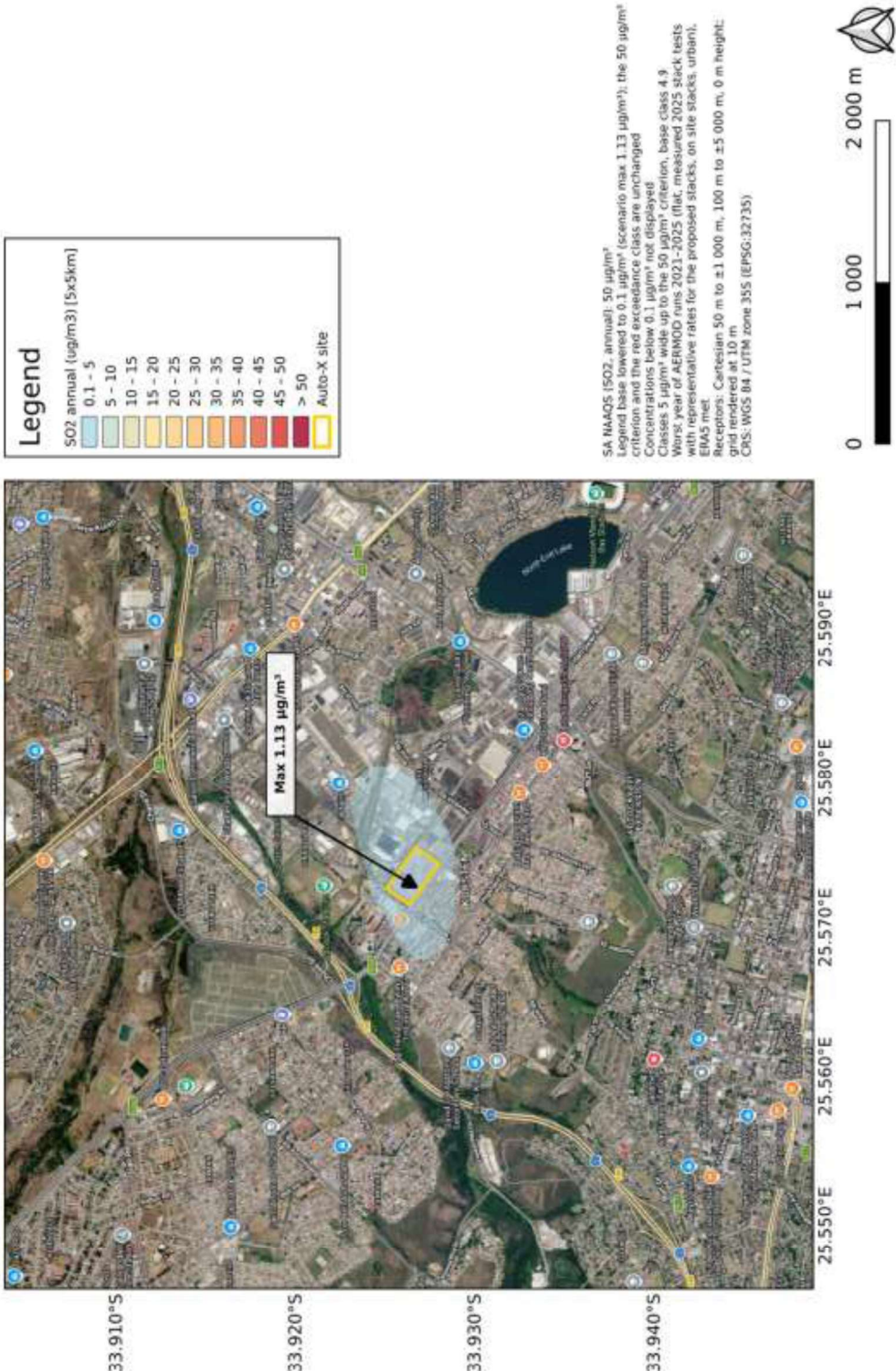


Figure 18: Scenario B — annual averaged SO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 50 µg/m³ in classes of 5 µg/m³; concentrations below 0.1 µg/m³ not displayed; map extent 5 km x 5 km.

NO2 — 1-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

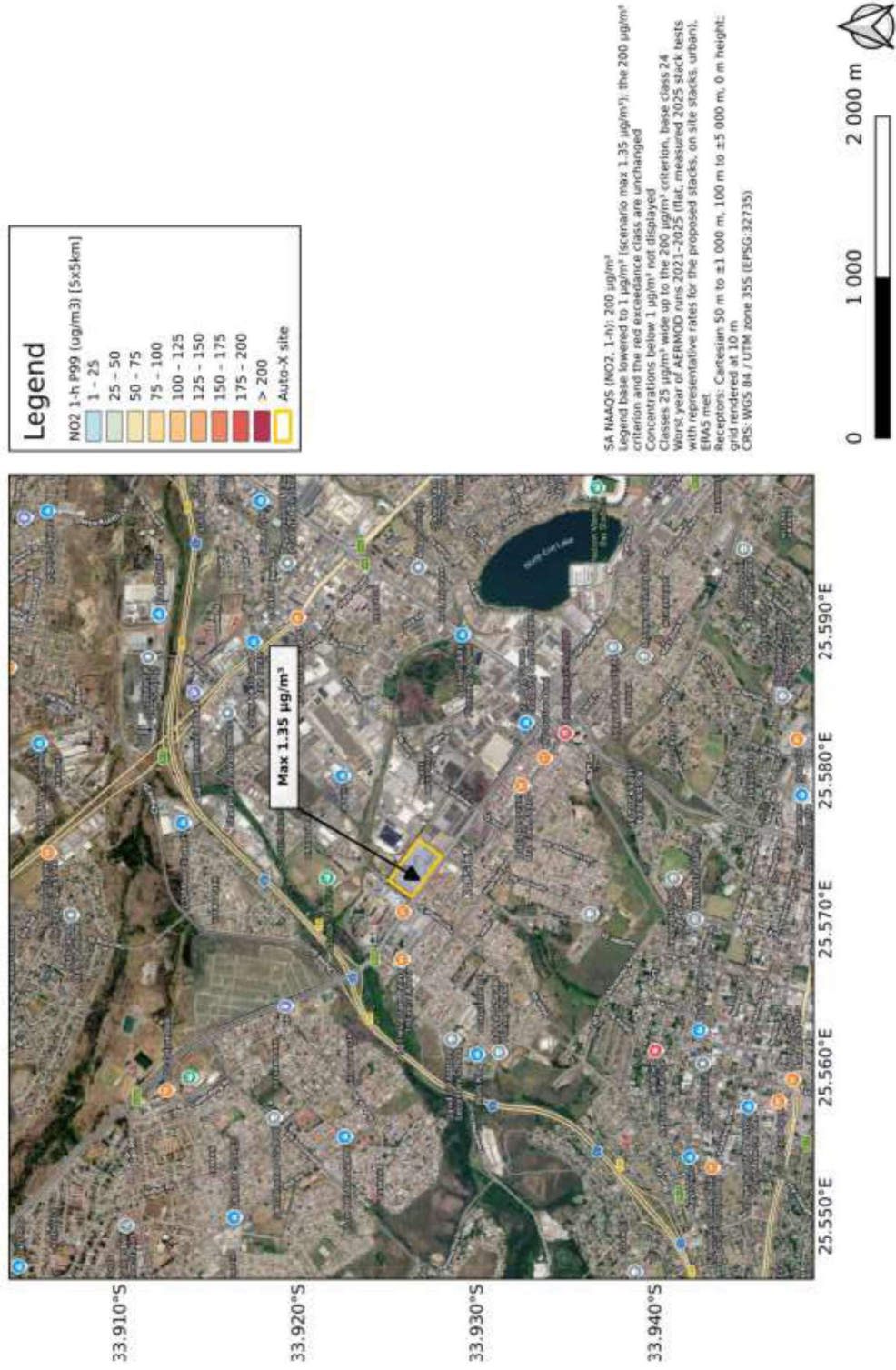


Figure 19: Scenario B — 99-percentile hourly NO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 200 µg/m³ in classes of 25 µg/m³; concentrations below 1 µg/m³ not displayed; map extent 5 km x 5 km.

NO2 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

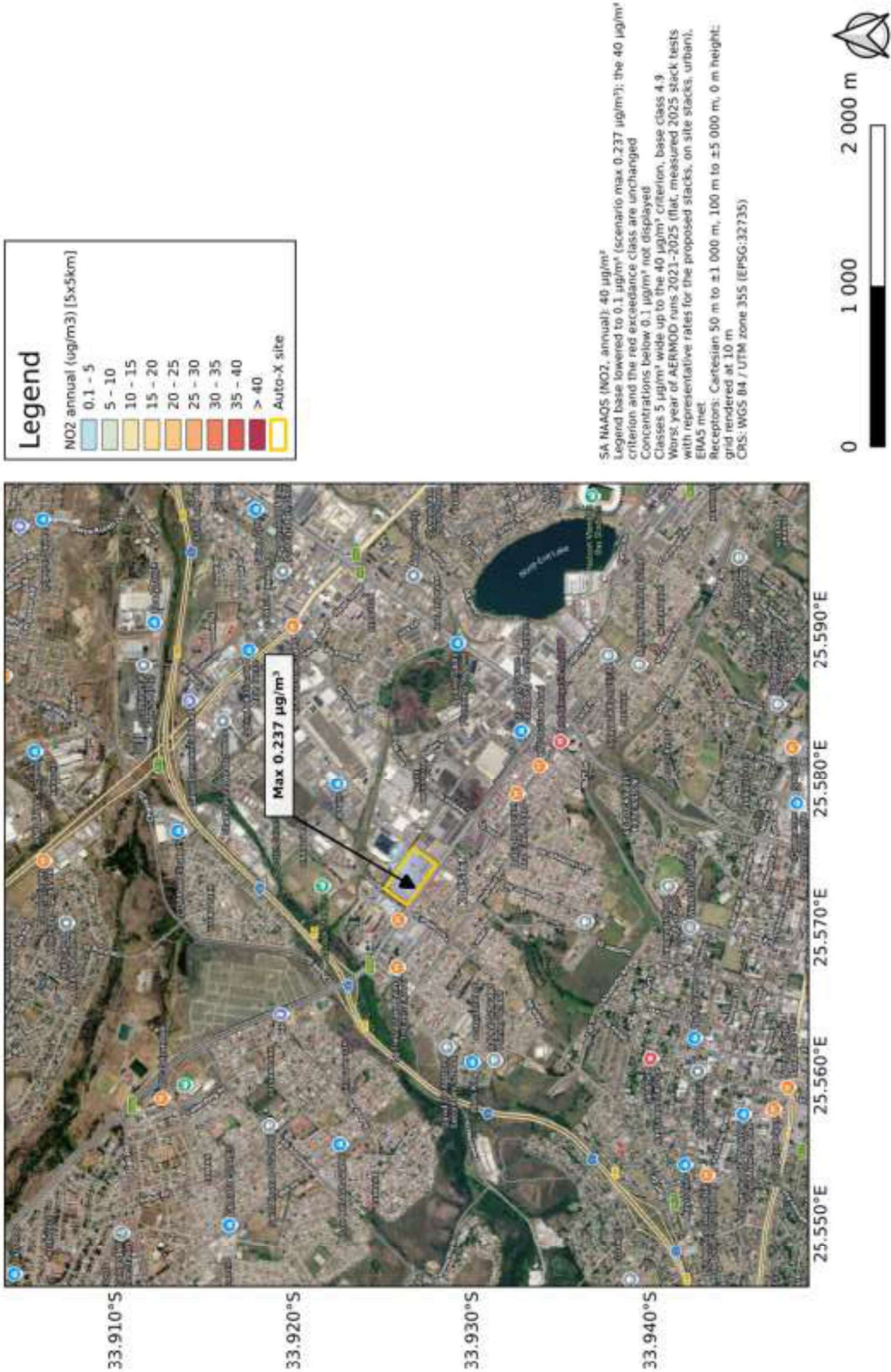


Figure 20: Scenario B — annual averaged NO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 40 µg/m³ in classes of 5 µg/m³; concentrations below 0.1 µg/m³ not displayed; map extent 5 km x 5 km.

H2SO4 — Maximum 1-hour average

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

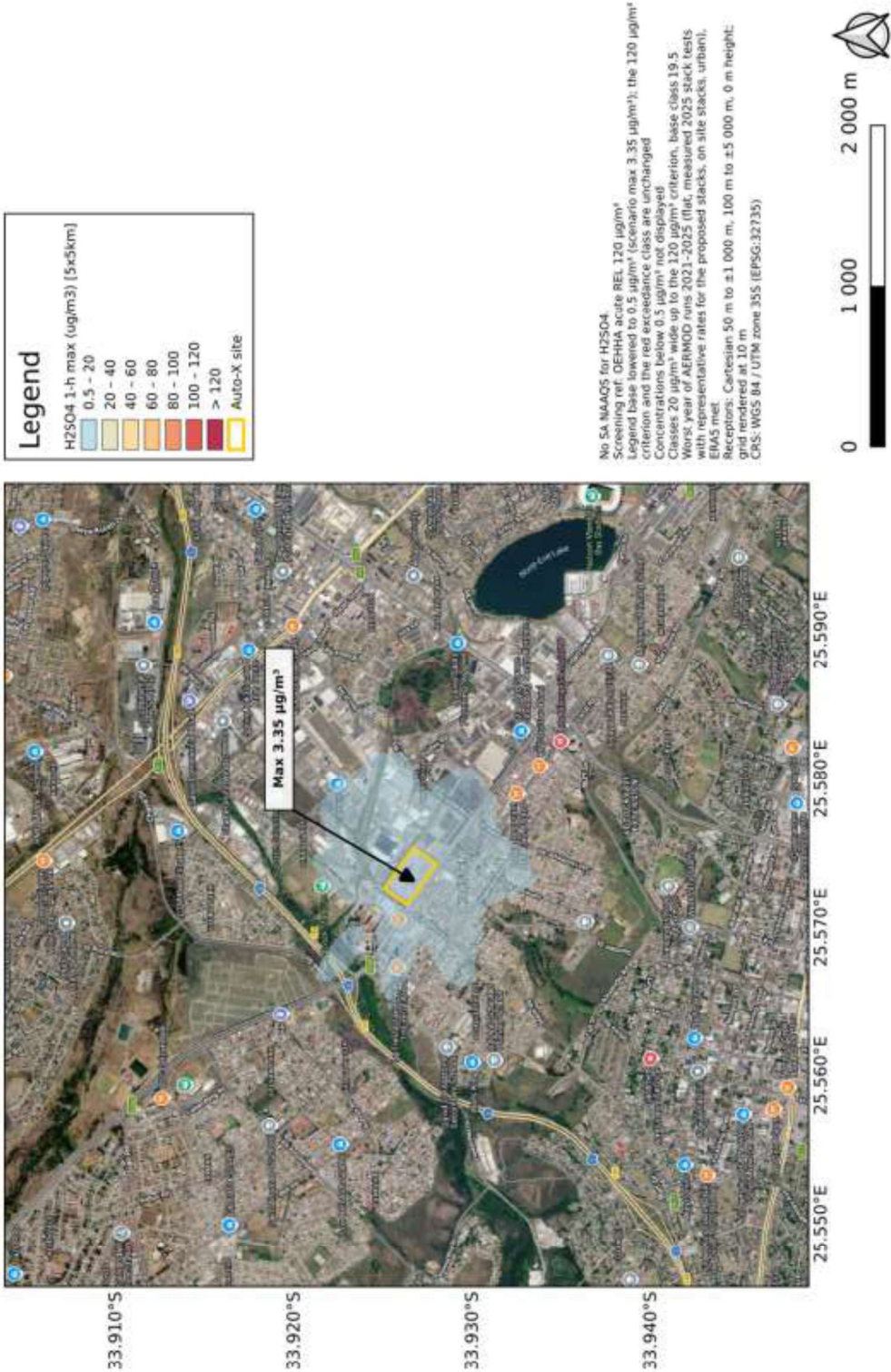


Figure 21: Scenario B — maximum 1-hour H2SO4 concentrations, µg/m³. Legend scaled to the OEHA acute REL of 120 µg/m³ in classes of 20 µg/m³; concentrations below 0.5 µg/m³ not displayed; map extent 5 km x 5 km.

H2SO4 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_SITE (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks)

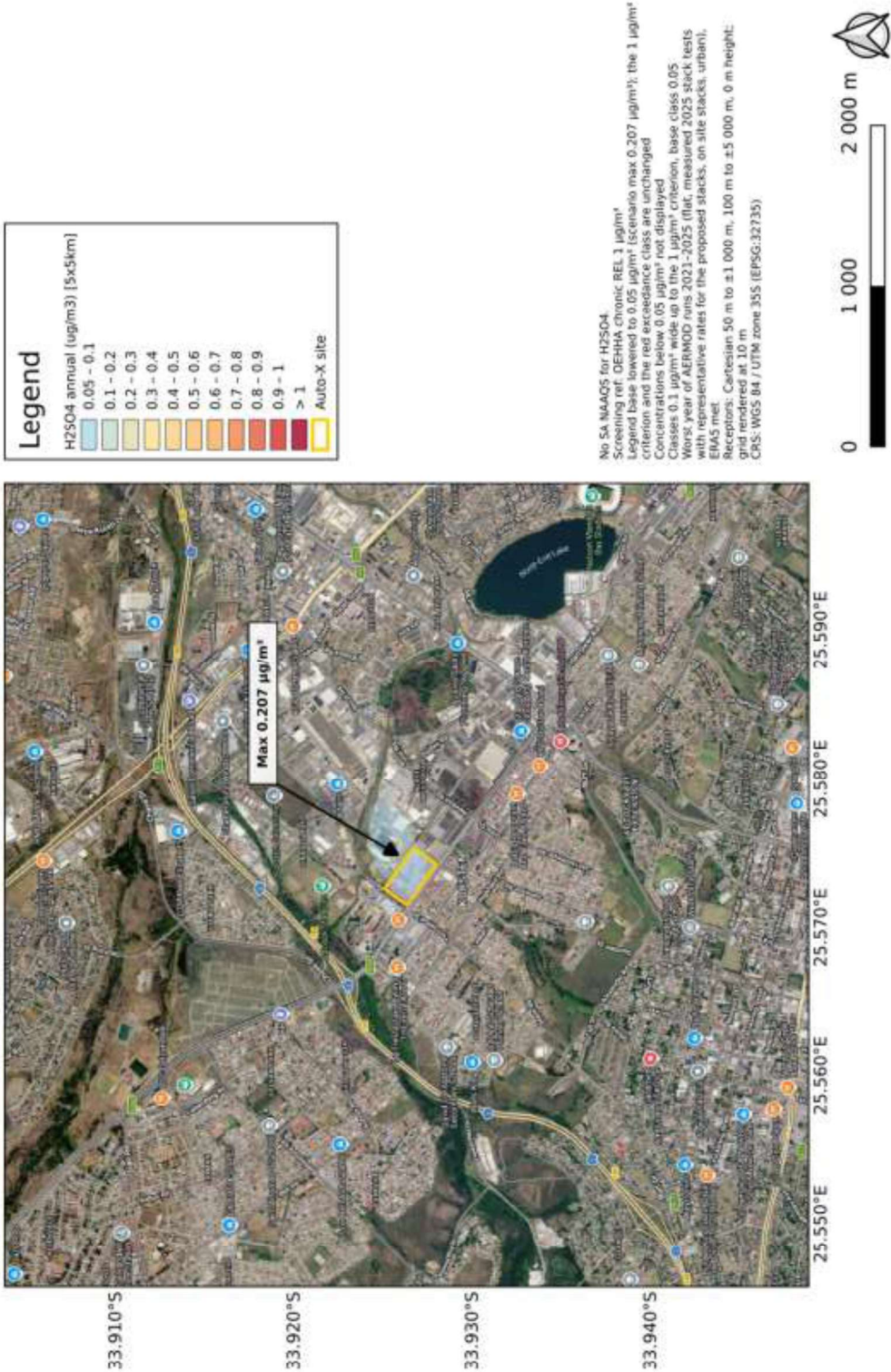


Figure 22: Scenario B — annual averaged H₂SO₄ concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the OEHHA chronic REL of 1 $\mu\text{g}/\text{m}^3$ in classes of 0.1 $\mu\text{g}/\text{m}^3$; concentrations below 0.05 $\mu\text{g}/\text{m}^3$ not displayed; map extent 5 km x 5 km.



7.3 Scenario C – cumulative impact

Cumulative impact, using the site emissions as measured for the existing sources, the estimated emission rates for the newly planned sources, and the actual emissions of the surrounding sources.

Pollutant / statistic	Limit (µg/m³)	Maximum (µg/m³)	Location of maximum from site centre	Maximum as % of limit	Maximum at a sensitive receptor (µg/m³)
PM ₁₀ , 24-hour 99th percentile	75	12.2	240 m SE	16%	3.5 (5%)
PM ₁₀ , annual average	40	5.4	150 m E	14%	1.4 (4%)
Pb, annual average	0.5	1.05	250 m ESE	211%	0.06 (13%)
SO ₂ , 1-hour 99th percentile	350	21.6	2.2 km ESE	6%	12.6 (4%)
SO ₂ , 24-hour 99th percentile	125	13.5	2.2 km ESE	11%	5.0 (4%)
SO ₂ , annual average	50	3.9	2.4 km ESE	8%	1.3 (3%)
NO ₂ , 1-hour 99th percentile	200	2.9	2.3 km ESE	1%	1.1 (0.6%)
NO ₂ , annual average	40	0.5	2.4 km ESE	1%	0.2 (0.4%)
H ₂ SO ₄ , maximum 1-hour average	120 (OEHHA acute REL, screening)	3.4	50 m ESE	3%	0.6 (0.5%)
H ₂ SO ₄ , annual average	1 (OEHHA chronic REL, screening)	0.21	230 m ENE	21%	0.01 (1%)

Table 9: Scenario C — predicted ground-level concentrations (worst year, 2021 – 2025).
Values above the applicable criterion are shown in red.

PM10 — 24-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

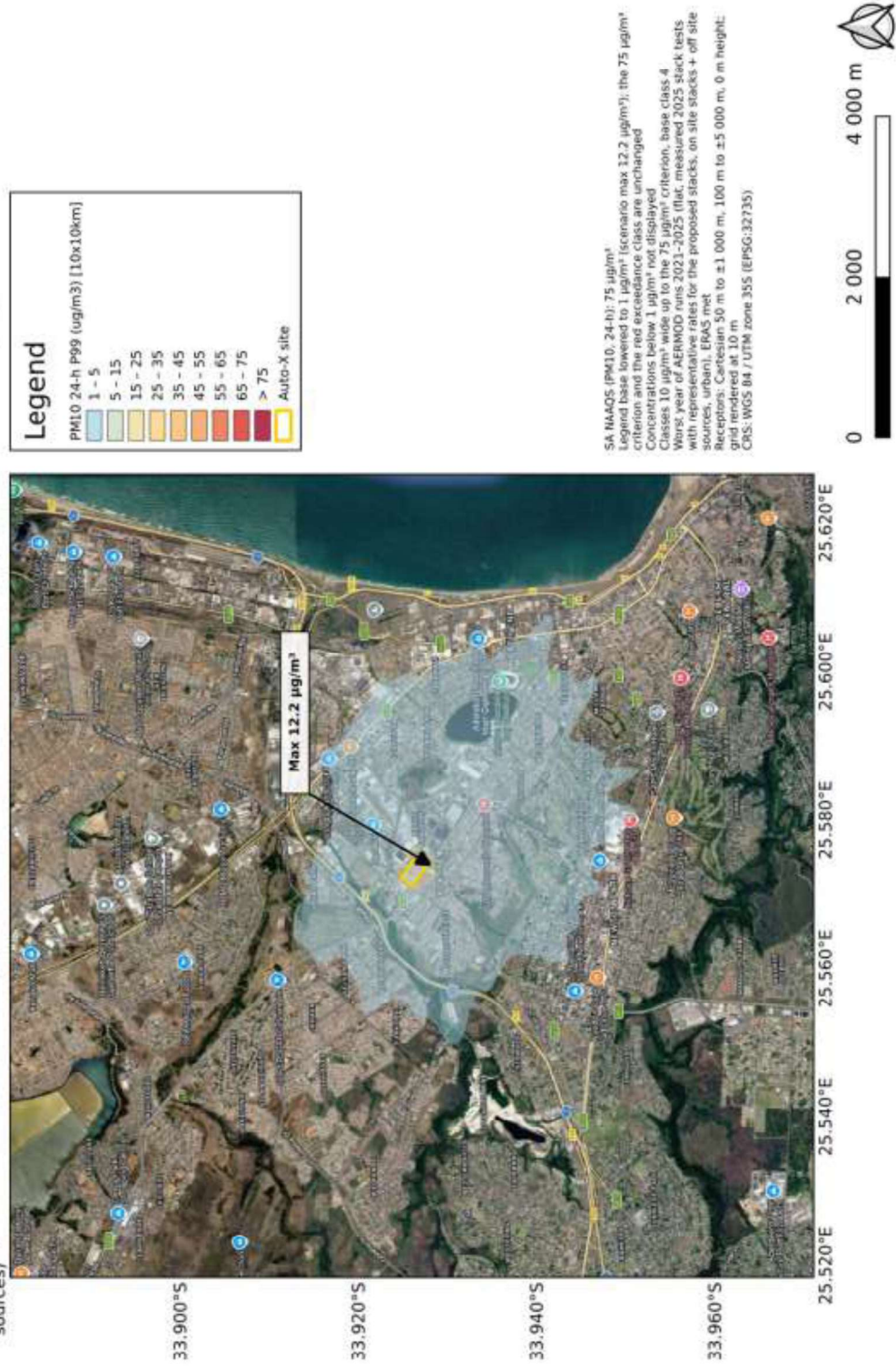


Figure 23: Scenario C — 99-percentile daily averaged PM₁₀ concentrations, µg/m³. Legend scaled to the NAAQS of 75 µg/m³ in classes of 5 µg/m³; concentrations below 1 µg/m³ not displayed; map extent 10 km x 10 km.

PM10 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

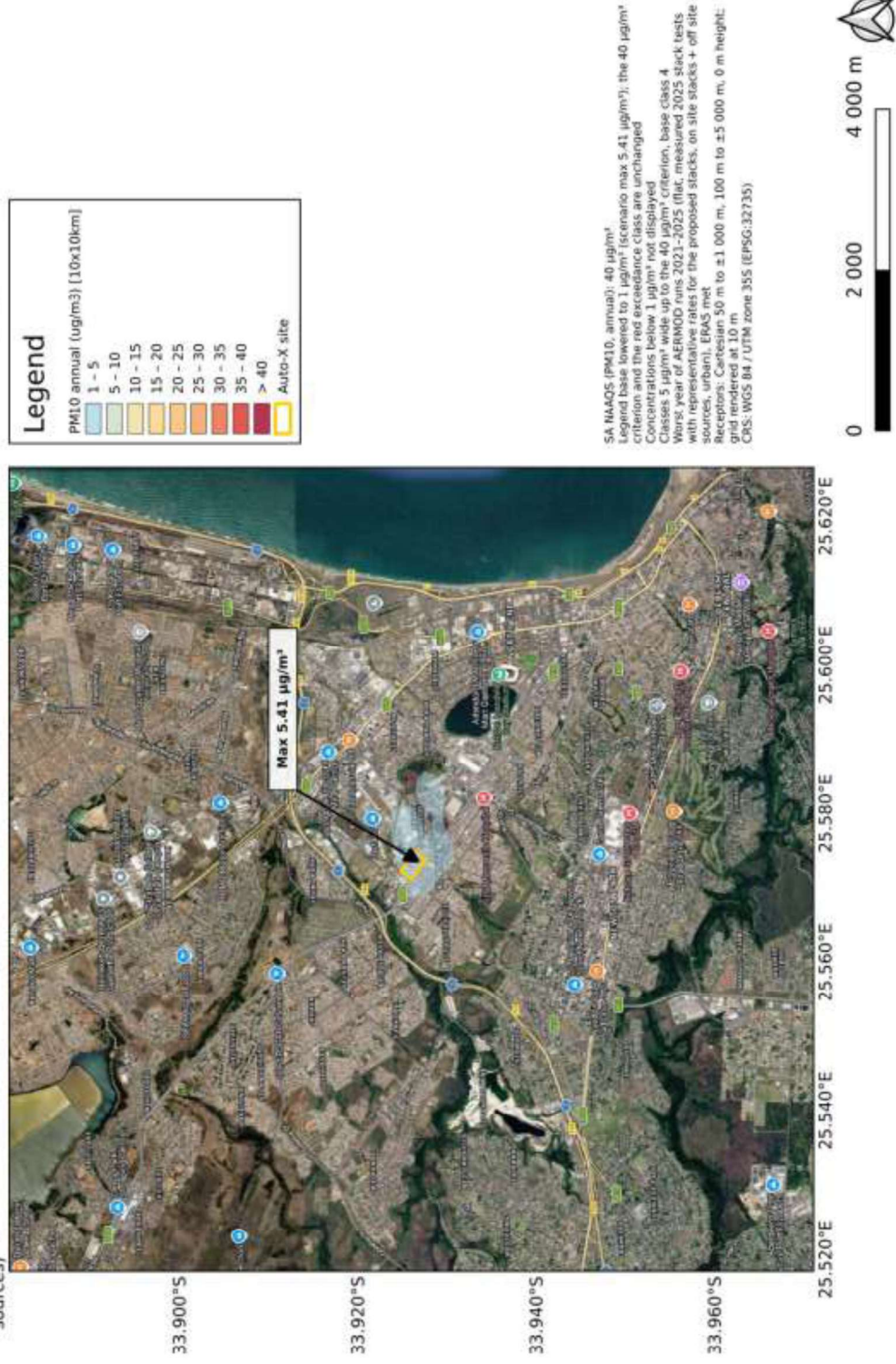


Figure 24: Scenario C — annual averaged PM₁₀ concentrations, µg/m³. Legend scaled to the NAAQS of 40 µg/m³ in classes of 5 µg/m³; concentrations below 1 µg/m³ not displayed; map extent 10 km x 10 km.

Pb — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

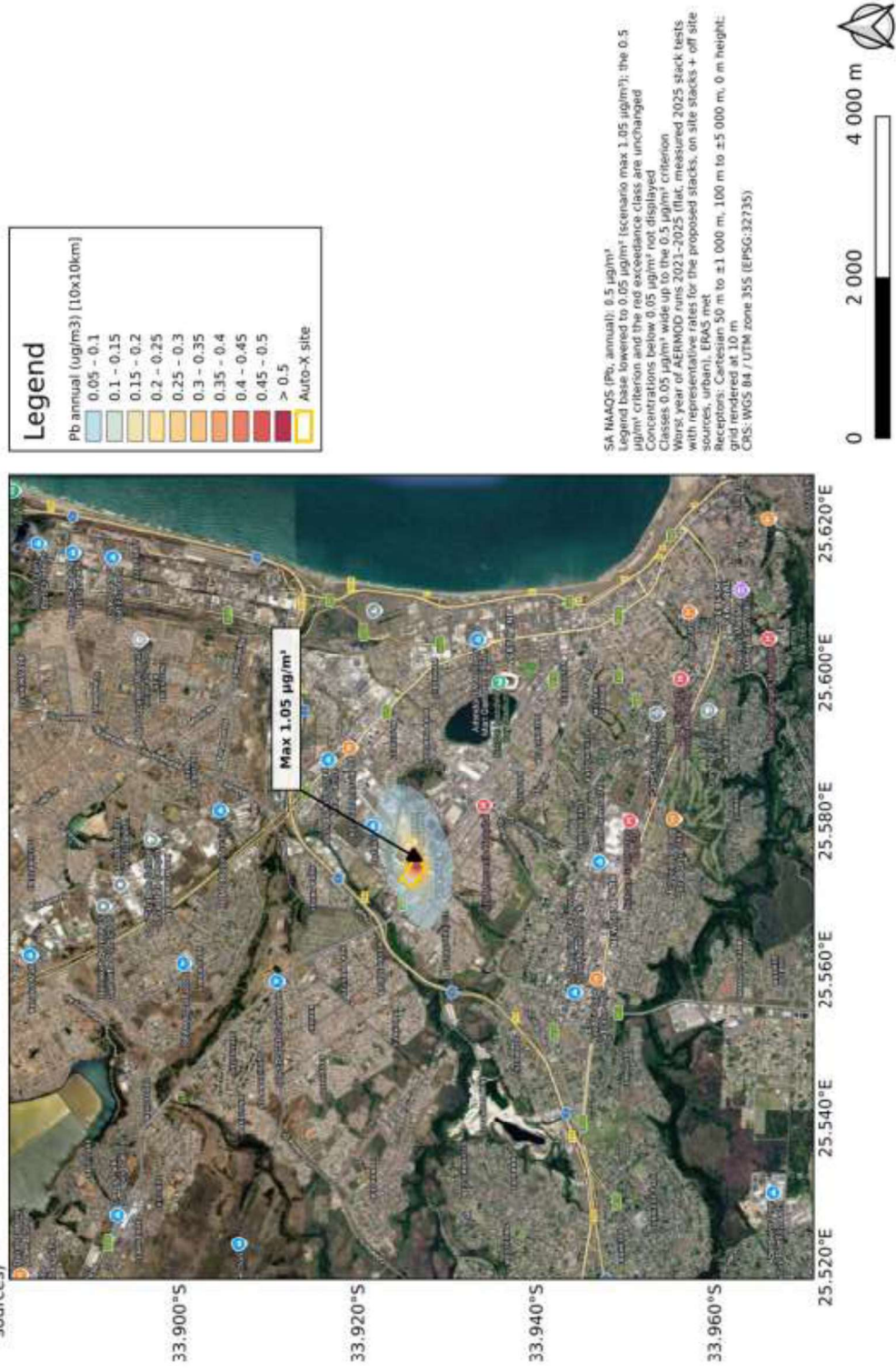


Figure 25: Scenario C — annual averaged Pb concentrations, µg/m³. Legend scaled to the NAAQS of 0.5 µg/m³ in classes of 0.05 µg/m³; concentrations below 0.05 µg/m³ not displayed; map extent 10 km x 10 km.

SO2 — 1-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

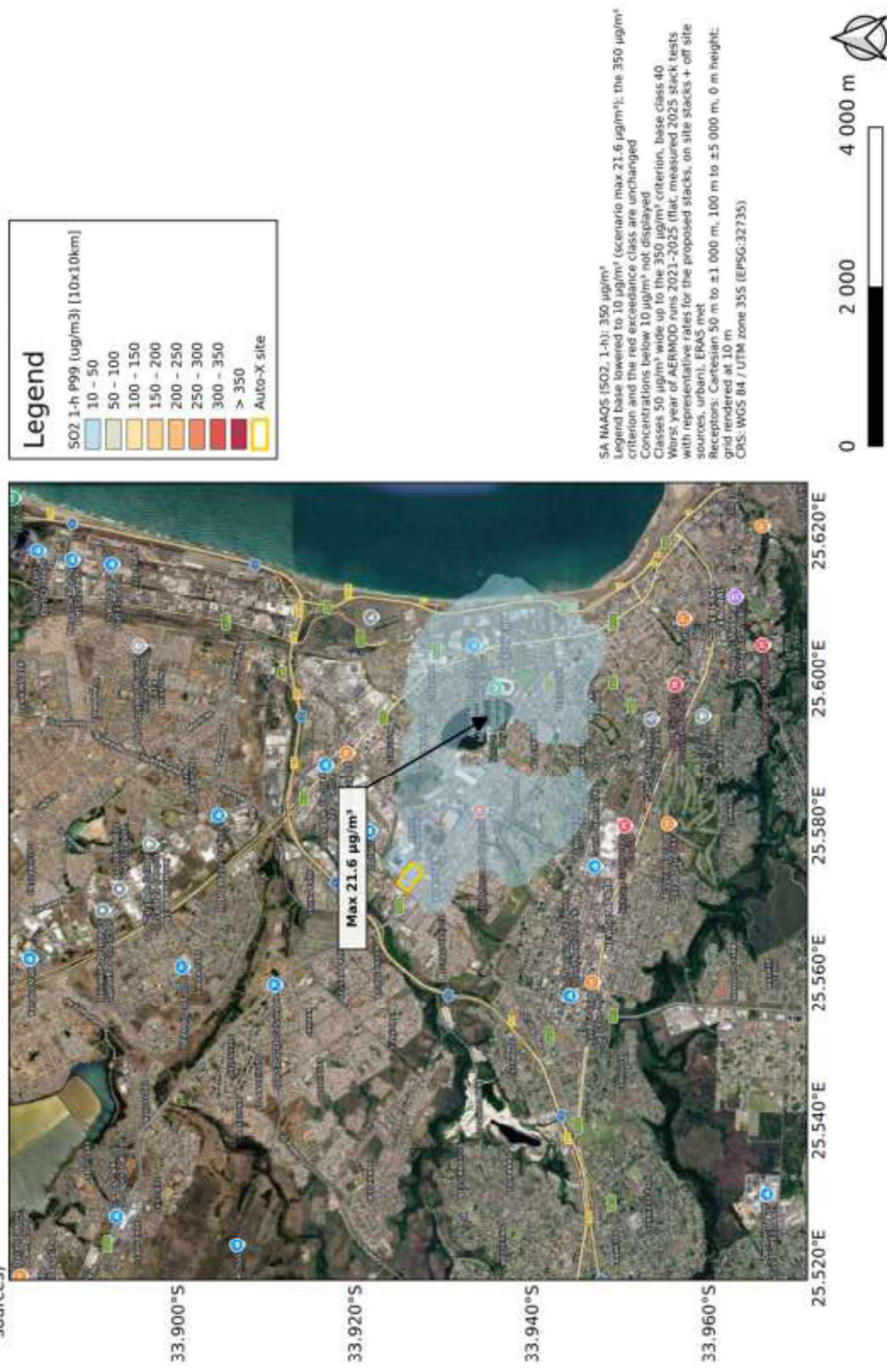


Figure 26: Scenario C — 99-percentile hourly SO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 350 µg/m³ in classes of 50 µg/m³; concentrations below 10 µg/m³ not displayed; map extent 10 km x 10 km.

SO2 — 24-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

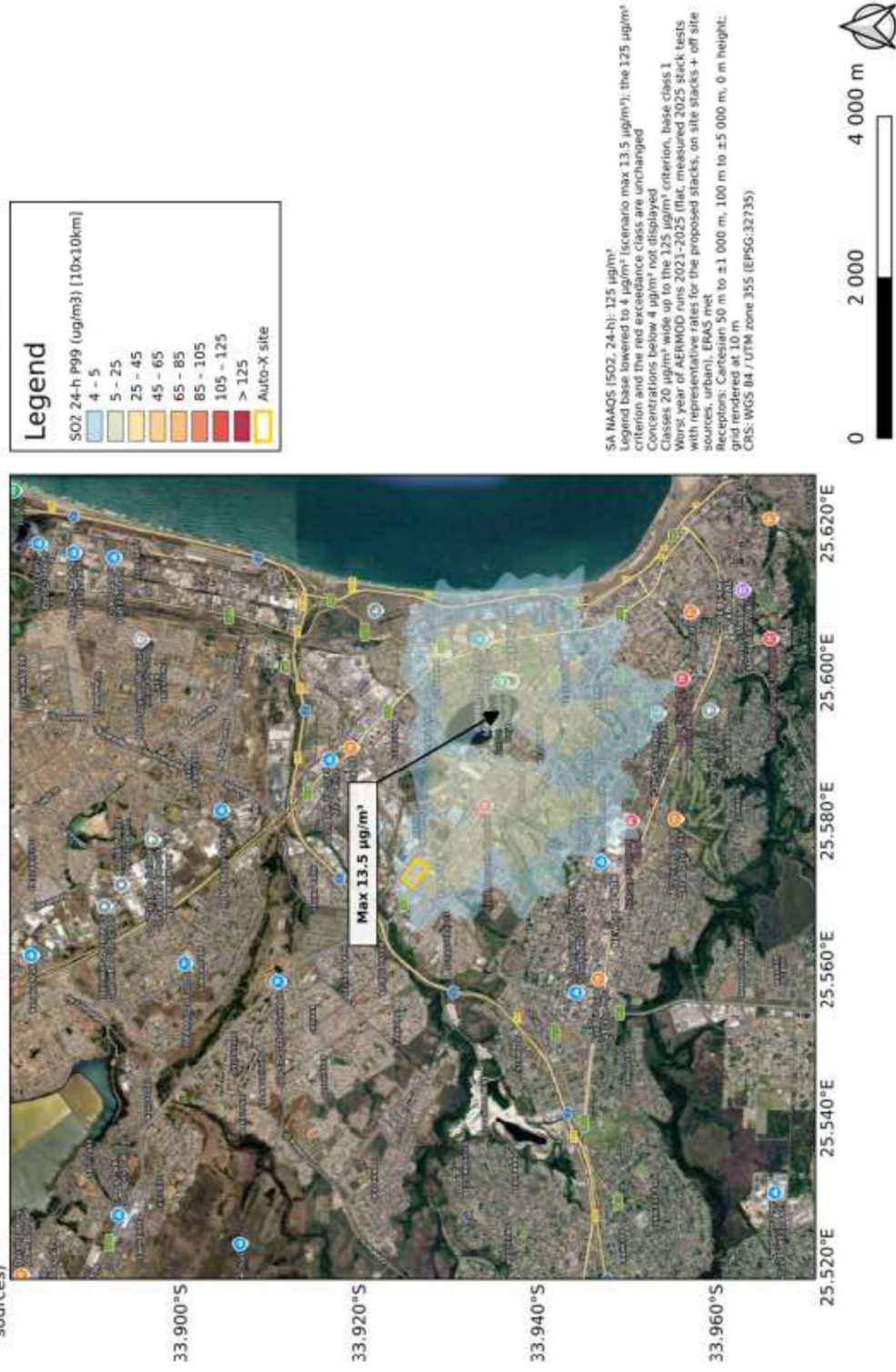


Figure 27: Scenario C — 99-percentile daily averaged SO₂ concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the NAAQS of 125 $\mu\text{g}/\text{m}^3$ in classes of 20 $\mu\text{g}/\text{m}^3$; concentrations below 4 $\mu\text{g}/\text{m}^3$ not displayed; map extent 10 km x 10 km.

SO2 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

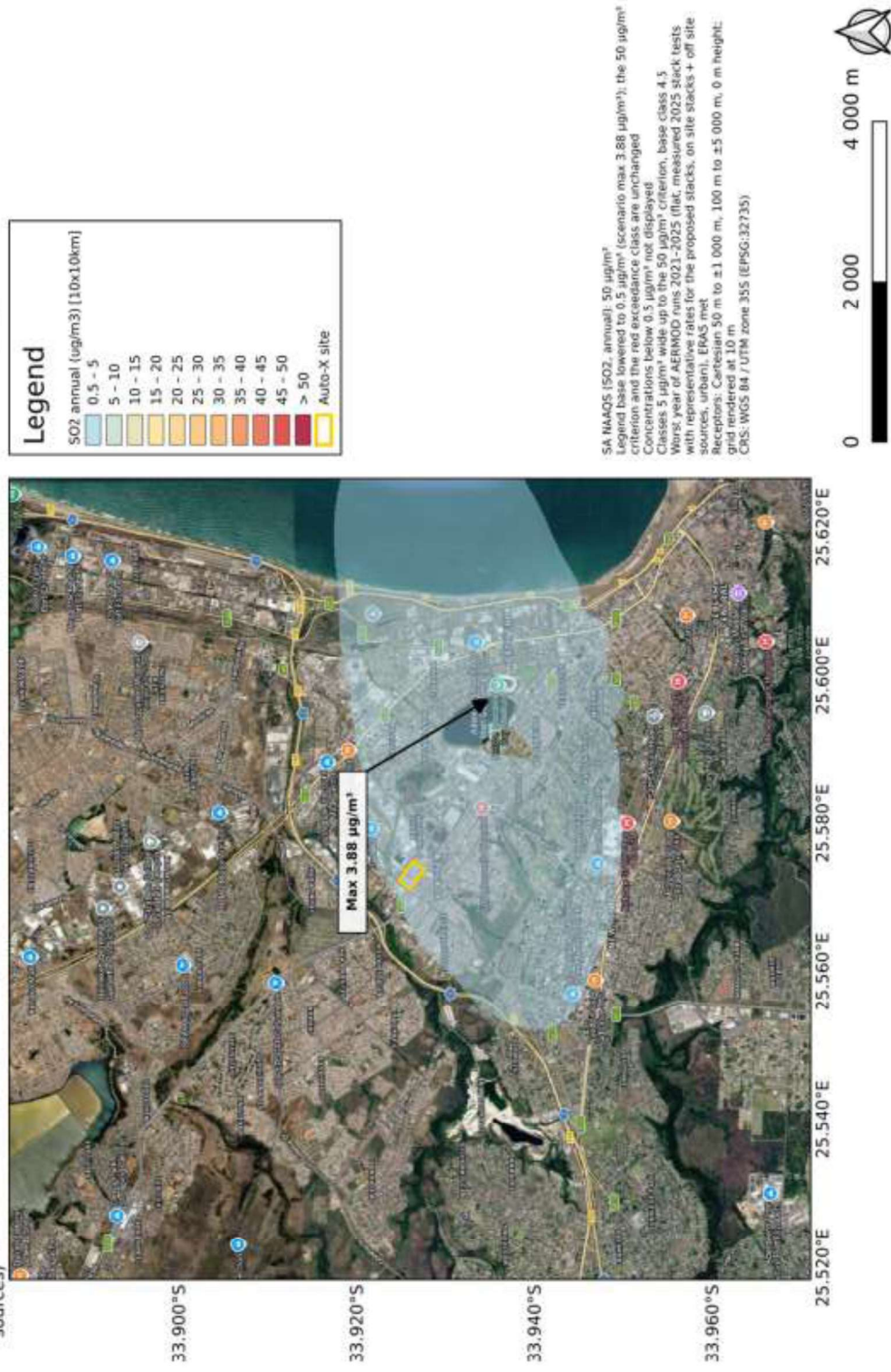


Figure 28: Scenario C — annual averaged SO₂ concentrations, $\mu\text{g}/\text{m}^3$. Legend scaled to the NAAQS of 50 $\mu\text{g}/\text{m}^3$ in classes of 5 $\mu\text{g}/\text{m}^3$; concentrations below 0.5 $\mu\text{g}/\text{m}^3$ not displayed; map extent 10 km x 10 km.

NO2 — 1-hour average, 99th percentile

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

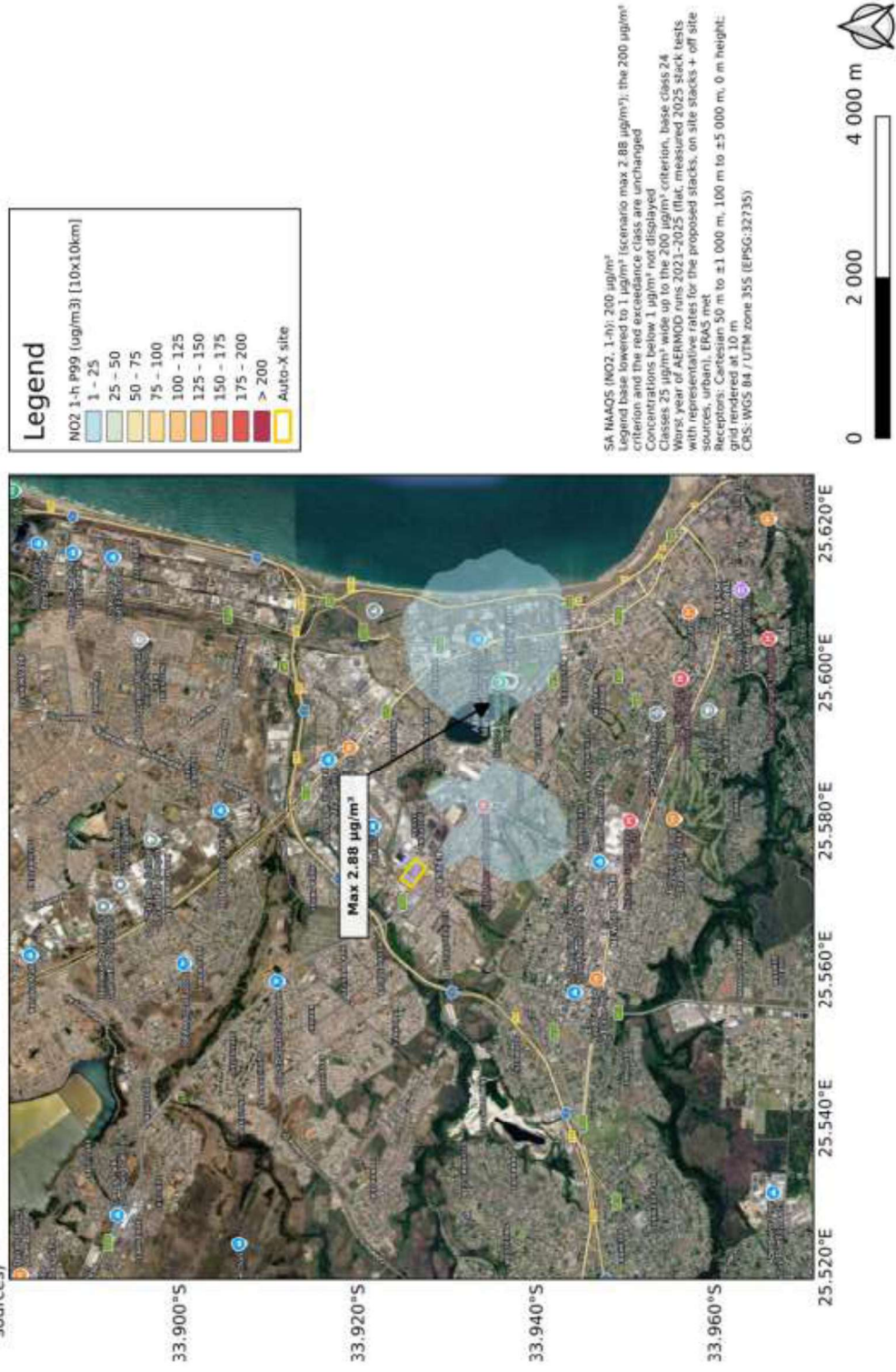


Figure 29: Scenario C — 99-percentile hourly NO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 200 µg/m³ in classes of 25 µg/m³; concentrations below 1 µg/m³ not displayed; map extent 10 km x 10 km.

NO2 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

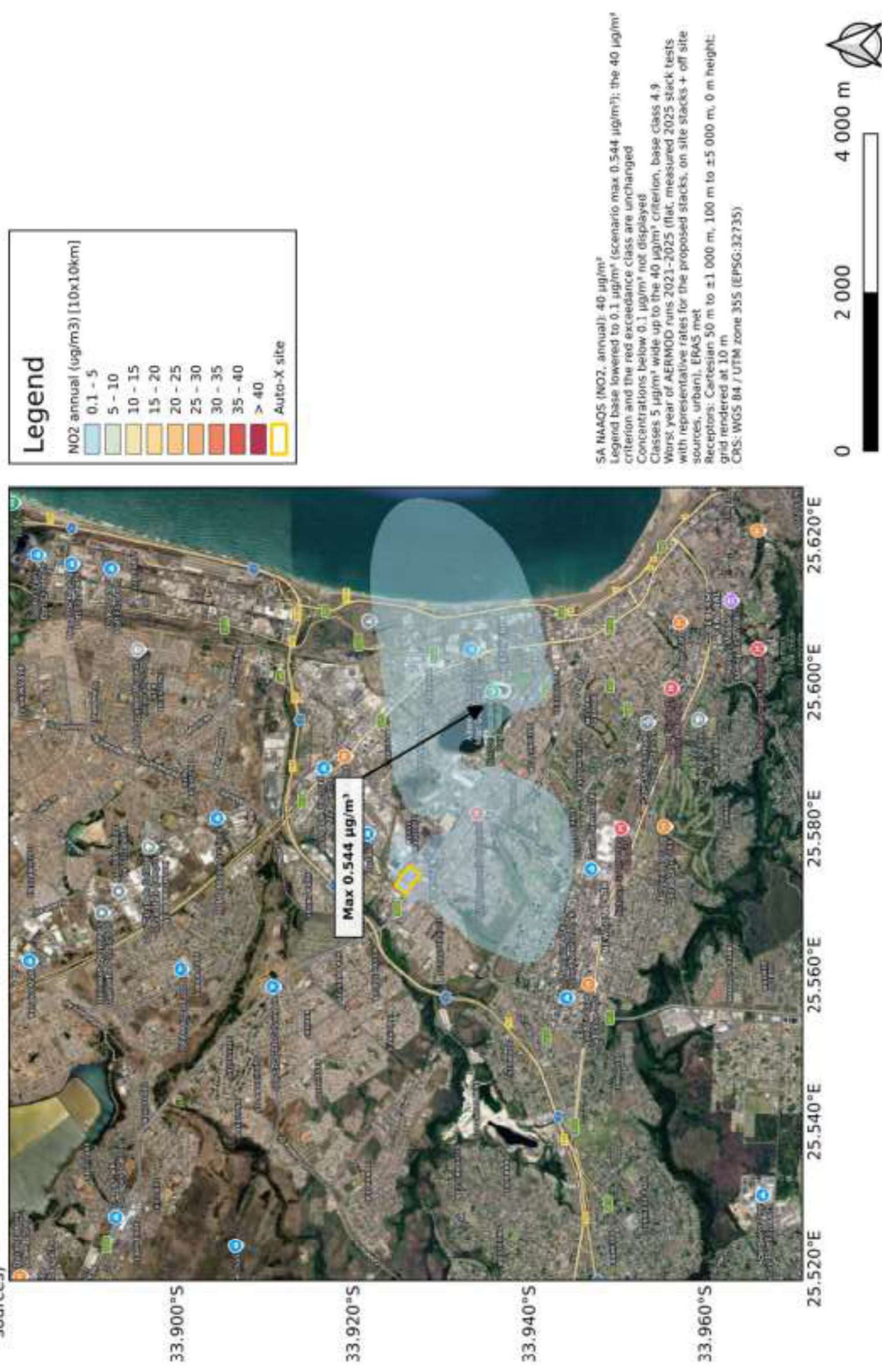


Figure 30: Scenario C — annual averaged NO₂ concentrations, µg/m³. Legend scaled to the NAAQS of 40 µg/m³ in classes of 5 µg/m³; concentrations below 0.1 µg/m³ not displayed; map extent 10 km x 10 km.

H2SO4 — Maximum 1-hour average

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

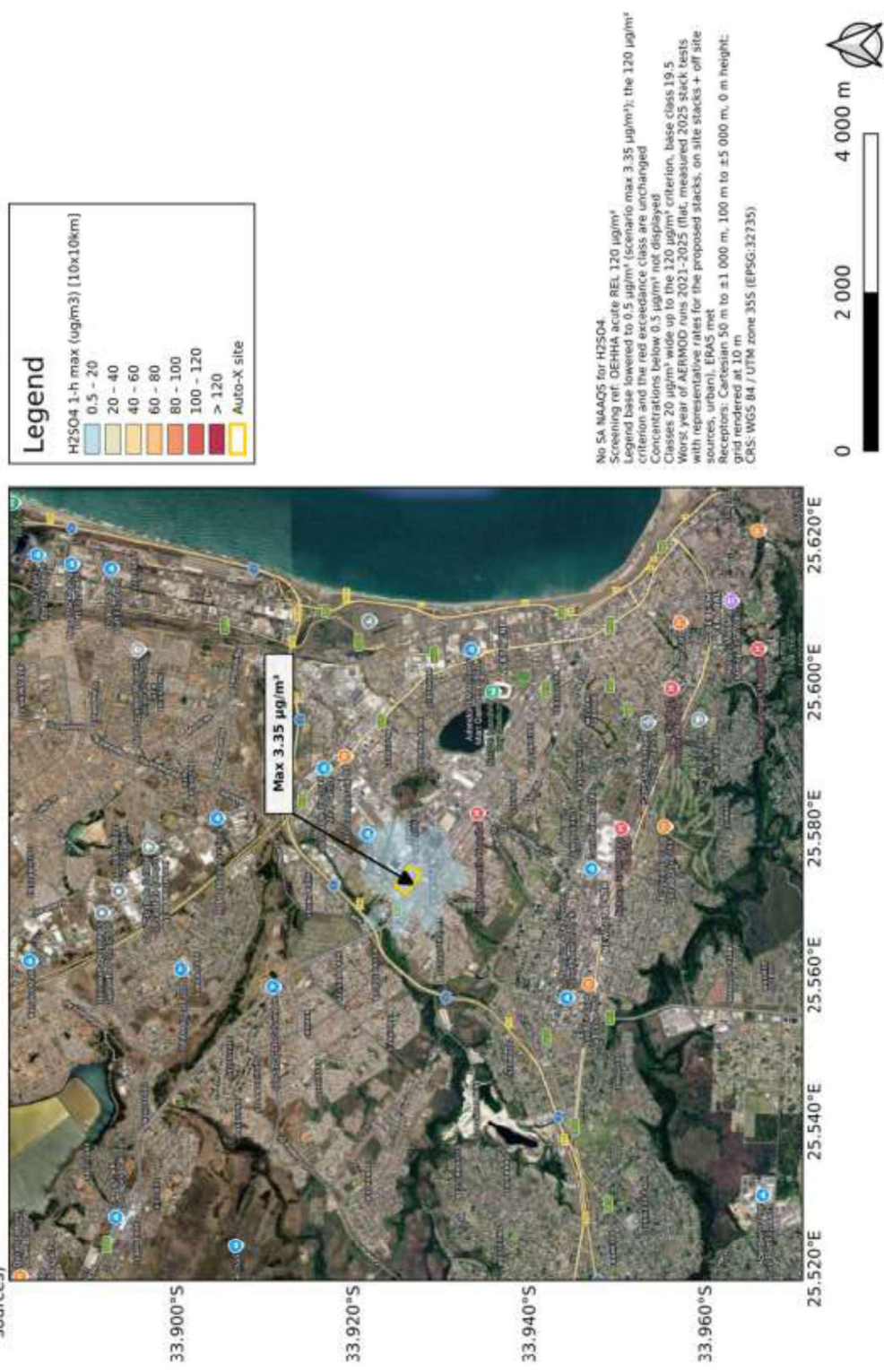


Figure 31: Scenario C — maximum 1-hour H₂SO₄ concentrations, µg/m³. Legend scaled to the OEHA acute REL of 120 µg/m³ in classes of 20 µg/m³; concentrations below 0.5 µg/m³ not displayed; map extent 10 km x 10 km.

H2SO4 — Annual average

Auto-X, Gqeberha — Model basis: MEASEST_CUMM (flat, measured 2025 stack tests with representative rates for the proposed stacks, on site stacks + off site sources)

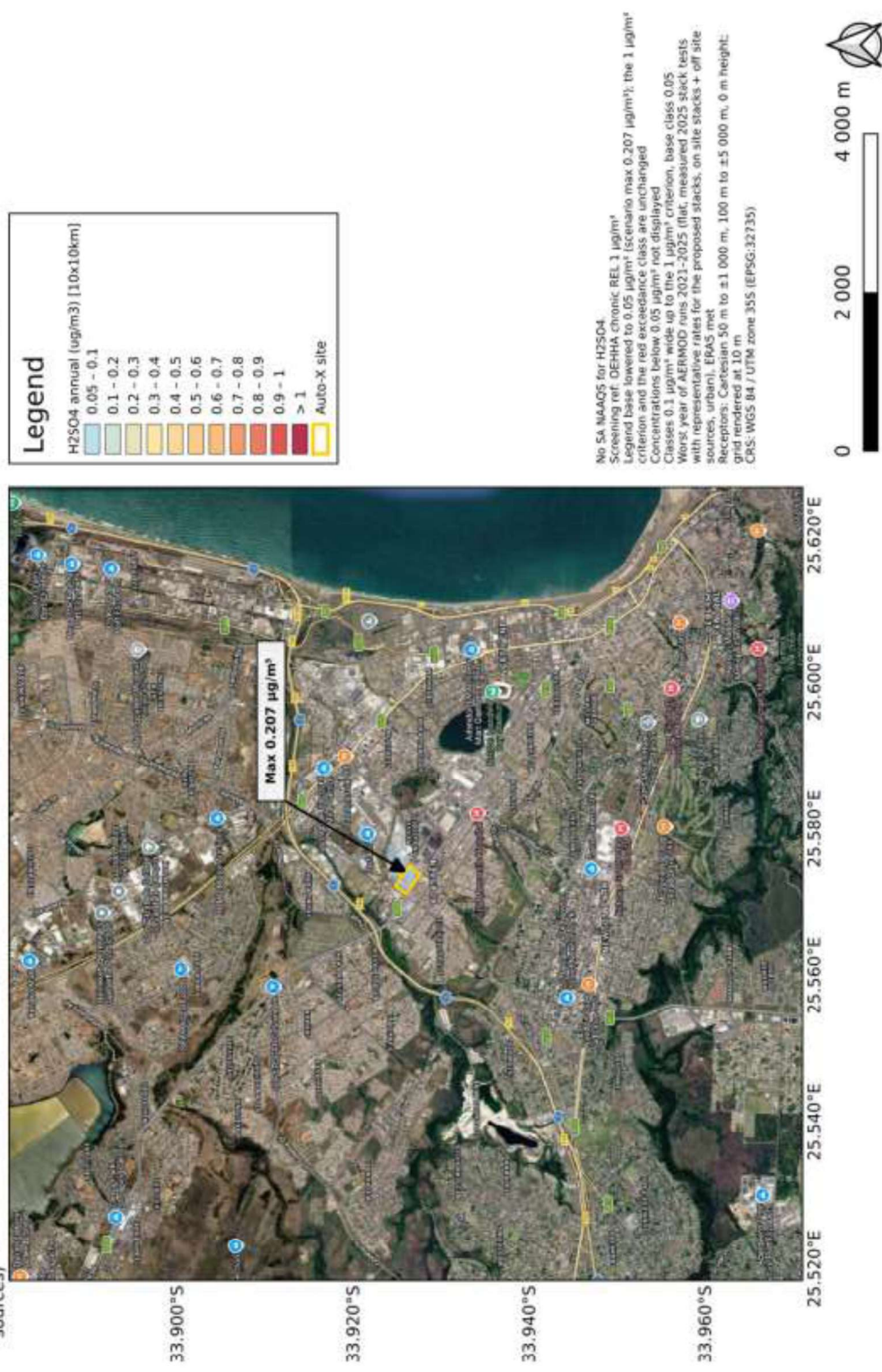


Figure 32: Scenario C — annual averaged H₂SO₄ concentrations, µg/m³. Legend scaled to the OEHHA chronic REL of 1 µg/m³ in classes of 0.1 µg/m³; concentrations below 0.05 µg/m³ not displayed; map extent 10 km x 10 km.



7.4 Comparison of the scenarios

All three scenarios were run on the same receptor grid, so they may be compared receptor by receptor and not only in their maxima.

Pollutant / statistic	Limit (µg/m³)	Scenario A (µg/m³)	Scenario B (µg/m³)	Scenario C (µg/m³)
PM ₁₀ , 24-hour 99th percentile	75	168	12.2	12.2
PM ₁₀ , annual average	40	77	5.22	5.41
Pb, annual average	0.5	5.10	1.05	1.05
SO ₂ , 1-hour 99th percentile	350	1 753	7.60	21.6
SO ₂ , 24-hour 99th percentile	125	792	4.15	13.5
SO ₂ , annual average	50	359	1.13	3.88
NO ₂ , 1-hour 99th percentile	200	1 196	1.35	2.88
NO ₂ , annual average	40	288	0.24	0.54
H ₂ SO ₄ , maximum 1-hour average	120 (OEHHA acute REL, screening)	274	3.35	3.35
H ₂ SO ₄ , annual average	1 (OEHHA chronic REL, screening)	31	0.21	0.21

Table 10: Maximum predicted ground-level concentration by scenario (worst year, 2021 – 2025). Values above the applicable criterion are shown in red.

The criteria exceeded on each scenario, at the point of maximum impact and at the sensitive receptors, are summarised in Table 11 below.



Scenario	Criteria exceeded at the point of maximum impact	Sensitive receptors at which a criterion is exceeded
Scenario A	every pollutant and averaging period assessed	7 of 16: Livingstone Hospital Primary School, Livingstone Hospital, Frank Joubert Primary School, De Vos Malan Primary School, Life Mercantile Private Hospital, Adolph Schauder Primary School and Abraham Levy Primary School
Scenario B	Lead (annual)	None
Scenario C	Lead (annual)	None

Table 11: Criteria exceeded, by scenario.

8 Impact on overall air quality

The results tabulated and mapped in Section 7 are assessed below, pollutant by pollutant and scenario by scenario, against the criteria set out in Section 6.

On every scenario associated with the operations of AutoX, the predicted maxima occur between 10 m and 260 m from site centre, that is either on or immediately beyond the plant boundary, where the short stacks and the building wake bring the plumes to ground. The nearest sensitive receptor is more than 350 m away. The concentrations predicted at the receptors are therefore substantially below the maxima tabulated in Section 7 and shown on the maps.

8.1 Particulate matter and lead

Scenario A – site emission only, based on emission limits

The maximum predicted 24-hour 99th-percentile PM₁₀ concentration is 168 µg/m³ located 220 m ESE of the site. This is substantially higher than the NAAQS of 75 µg/m³. The maximum predicted annual average is 77 µg/m³ located 160 m ESE of the site. This also exceeds the NAAQS of 40 µg/m³.

The maximum predicted annual average lead concentration is 5.10 µg/m³ occurring approximately 160 m ESE of site centre. This exceeds the NAAQS of 0.5 µg/m³ by a full order of magnitude.

At the nearest sensitive receptor, the Frank Joubert Primary School, the highest predicted 24-hour 99th-percentile PM₁₀ concentration is 41 µg/m³ and the highest annual average PM₁₀ concentration is 6.91 µg/m³, well below the air quality standard.

For lead the highest value recorded at the school is 0.46 µg/m³, i.e. marginally below the air quality standard of 0.5 µg/m³.



Scenario B – site emissions only, based on actual emissions

The maximum predicted 24-hour 99th-percentile PM₁₀ concentration is 12.2 µg/m³ located 240 m SE of the site. This is well below the NAAQS of 75 µg/m³. The maximum predicted annual average is 5.2 µg/m³ located 150 m E of the site. This is also well below the NAAQS of 40 µg/m³.

The maximum predicted annual average lead concentration is 1.05 µg/m³ occurring approximately 250 m ESE of site centre. This exceeds the NAAQS of 0.5 µg/m³. Lead is the only pollutant for which a standard is exceeded on this basis.

At the sensitive receptors the highest predicted 24-hour 99th-percentile PM₁₀ concentration is 2.8 µg/m³ and the highest annual average PM₁₀ concentration is 0.5 µg/m³, both well below the air quality standards.

For lead, the highest value recorded at a sensitive receptor is 0.06 µg/m³ at the Frank Joubert Primary School. This is well below the air quality standard of 0.5 µg/m³.

Scenario C – cumulative impact based on actual emissions

The maximum predicted 24-hour 99th-percentile PM₁₀ concentration remains 12.2 µg/m³ located 240 m SE of the site. The maximum predicted annual average increases marginally to 5.4 µg/m³ located 150 m E of the site. This is still well below the NAAQS of 40 µg/m³.

At the sensitive receptors the highest predicted 24-hour 99th-percentile PM₁₀ concentration is 3.5 µg/m³ and the highest annual average PM₁₀ concentration is 1.4 µg/m³, both well below the air quality standards.

The maximum predicted annual average lead concentration remains unchanged from Scenario B.

8.2 Sulphur dioxide and nitrogen dioxide

Scenario A

The maximum predicted 1-hour 99th-percentile SO₂ concentration is 1 753 µg/m³ exceeding the NAAQS value of 350 µg/m³. The maximum value at the worst affected sensitive receptor, i.e. the Livingstone Hospital, is below the NAAQS value at 325 µg/m³.

The maximum 24-hour 99th percentile is 792 µg/m³ at the point of maximum impact and 170 µg/m³ at the most affected sensitive receptor. This exceeds the standard of 125 µg/m³ both near the site and at the worst affected sensitive receptor. The standard is exceeded at the Livingstone Hospital, the Livingstone Hospital Primary School and the Frank Joubert Primary School.

The 1-hour 99th percentile NO₂ value reaches a maximum of 1 196 µg/m³ 50 m ENE of the site, which exceed the NAAQS of 200 µg/m³. It is however below the standard at the nearest receptor at a concentration of 148 µg/m³.

The maximum predicted annual average concentration of SO₂ and NO₂ is 359 and 288 µg/m³, respectively. Both maximum annual averages substantially exceed the NAAQS concentrations of 50 µg/m³ for SO₂ and 40 µg/m³ for NO₂. For SO₂ and NO₂, the annual average concentrations determined at the worst affected sensitive receptors are below the NAAQS concentration.



Scenario B

Scenario B shows that the impact of AutoX's operations on the ground level concentrations of SO₂ and NO₂ is practically negligible in the context of this report. This is expected as the processes used by AutoX do not generate SO₂ or NO₂ emissions in significant quantities.

The maximum predicted 1-hour 99th-percentile SO₂ and NO₂ concentrations are 7.6 µg/m³ and 1.4 µg/m³, respectively. This is well below the applicable NAAQS of 350 and 200 µg/m³.

For SO₂, the maximum 24-hour 99th percentile is 4.2 µg/m³ which is well below the NAAQS of 125 µg/m³.

The maximum predicted annual average for SO₂ and NO₂ is 1.1 and 0.2 µg/m³, respectively. This well below the corresponding NAAQS values of 50 and 40 µg/m³.

At all sensitive receptors the values are even lower than these maximums.

Scenario C

The maximum predicted 1-hour 99th-percentile SO₂ concentration is 21.6 µg/m³, well below the 350 µg/m³ NAAQS. The maximum predicted 24-hour 99th percentile is 13.5 µg/m³, well below the 125 µg/m³ NAAQS. The maximum annual average SO₂ is 3.9 µg/m³, well below the 50 µg/m³ NAAQS.

The maximum 1-hour 99th-percentile NO₂ reaches an overall maximum of 2.9 µg/m³, well below the 200 µg/m³ NAAQS. The annual average reaches a maximum of 0.5 µg/m³, also well below the standard of 40 µg/m³,

These maxima no longer occur at the site boundary but lie 2.2 km to 2.4 km ESE of site centre as the primary sources of these pollutants are external.

Overall

For Scenario A the 24-hour SO₂ standard is exceeded at three sensitive receptors — Livingstone Hospital Primary School (170 µg/m³), Livingstone Hospital (170 µg/m³) and Frank Joubert Primary School (133 µg/m³) — 350 m to 470 m from site centre.

For Scenarios B and C no sulphur dioxide or nitrogen dioxide standard is exceeded at any sensitive receptor, at the point of maximum impact, or anywhere in the modelled domain.

These results must be read with Section 5.2.3. The greater part of the modelled SO₂ and NO_x is listed against two stacks that are not expected to emit either pollutant, and the 2025 stack survey measured both at or below the detection limit at those stacks. The predicted SO₂ and NO₂ concentrations on the licensed basis are only a result of how the AEL schedule allocates emission limits and how these limits should be modelled in an AQI. They are however not a prediction of what the plant releases, which Scenario B shows to be much less.



8.3 Sulphuric acid

There are no South African ambient air quality standards for sulphuric acid. The occupational threshold limit value (TLV-TWA) applied in workplace health applications is $200 \mu\text{g}/\text{m}^3$, being the concentration to which a healthy adult worker may be exposed for eight hours per day. The occupational limit is quoted for context only and is not an ambient criterion.

For the assessment of ambient exposure, the OEHHA Reference Exposure Levels are applied: a chronic REL of $1 \mu\text{g}/\text{m}^3$ as an annual average, and an acute REL of $120 \mu\text{g}/\text{m}^3$ as a 1-hour average, both derived for the protection of the general population against respiratory effects.

Scenario A

The maximum predicted annual average concentration is $31 \mu\text{g}/\text{m}^3$, occurring approximately 50 m ESE of site centre; the highest value at a sensitive receptor is $1.99 \mu\text{g}/\text{m}^3$. This substantially exceeds the chronic REL of $1 \mu\text{g}/\text{m}^3$. As can be seen from Table 7, the chronic REL is exceeded at 7 of 16 receptors. The maximum predicted 1-hour concentration is $274 \mu\text{g}/\text{m}^3$ which also exceeds the acute REL of $120 \mu\text{g}/\text{m}^3$, however the acute REL is not exceeded at any of the listed sensitive receptors.

Scenario B

When considering only the actual emissions from the plant, the results change significantly. The maximum predicted annual average concentration is $0.2 \mu\text{g}/\text{m}^3$, occurring approximately 230 m ENE of site centre and the highest value at a sensitive receptor is $0.01 \mu\text{g}/\text{m}^3$. The maximum predicted 1-hour concentration is $3.4 \mu\text{g}/\text{m}^3$. Neither reference exposure level is exceeded at the point of maximum impact or at any receptor.

Scenario C

These values are identical to Scenario B, as no surrounding source emits sulphuric acid mist.

These are screening references and gazetted standards. An exceedance identifies a pollutant that warrants further consideration; it is not a breach of a South African ambient air quality standard and must not be reported as one.



9 Discussion

9.1 Model reliability

The results of any computer model are only as reliable as the quality of the input data.

9.1.1 Emissions

All particulate emissions were regarded as PM₁₀, while PM₁₀ particulates are actually a subset of total particulate matter emissions. Emissions were modelled as continuous (24 hours per day, 365 days per year) although actual uptime of the plant will be less due to maintenance shutdowns and downtime.

A degree of uncertainty exists in all calculations due to the following reasons:

- For Scenario C, there are no measured data for the proposed expansion, hence the same emission basis as for Scenario B is used for these stacks.
- Emission test results, from which data such as stack flow rates and temperatures are obtained, represent only a small fraction of annual operating hours. Typically, a set of emission measurements consists of three one-hour tests while a typical annual operating cycle consists of several thousand hours. Extrapolating from such a small sample introduces uncertainty.
- For Scenario A, emissions were derived from the applicable minimum emission standards rather than from measured stack concentrations, so the modelled rates represent the maximum permitted emission rather than the plant's actual emission. This is conservative wherever the plant operates below its permitted limits.
- For scenario A the SO₂ and NO_x emissions listed against the charge room extraction stacks are not expected to arise there (Section 5.2.3). They are modelled as listed because the Regulations require assessment at the maximum permitted concentration and the predicted SO₂ and NO₂ fields are a consequence of the licence emission limits rather than a prediction of the plant's actual impact.
- Scenario B is based on a stack-testing campaign for the plant, and typical (expected) emissions from the new stacks.
- Scenario C is based on the emissions data used for Scenario B, as well as data extracted from LAQS' emission database. The data is not necessarily current or representative as some of the data may be outdated due to, e.g., changes on operating conditions at the various industries involved.
- The cumulative impact assessment does not consider all contributing sources, only those for which data was readily available. Other sources will contribute to ambient concentrations. These include particulate and NO_x emissions from vehicles, as well as possible lead emissions from fossil fuel burning installations not already considered.



9.1.2 Meteorological and terrain data

The ERA5 reanalysis dataset assembled by LAQS covers five full consecutive years (2021 – 2025) and no gaps exist in the dataset. It includes all of the parameters required for dispersion modelling purposes. Although the dataset consists of simulated data, it assimilates observations and is regarded as the most reliable meteorological basis available for the study area in the absence of a representative surface station. The use of simulated data is an acceptable practice according to GN R.533. Modelling each year separately and reporting the worst year adds further conservatism.

As with any simulated dataset some degree of uncertainty exists when compared to measured data as generalised conditions of the area apply to the simulated data which may differ at the exact site location. These effects are considered minor, and the dataset is regarded as suitable for the purposes of this study.

9.1.3 Terrain and building data

The reported runs were carried out on a flat-terrain basis as it was determined that this is a conservative choice with respect to the predicted maximum ground-level concentration. As the plant's stacks are low (4.0 m to 8.5 m) relative to the surroundings, the dispersion of pollutants is affected by nearby structures. Although the model accounts for these effects, another uncertainty is introduced relating to the description of nearby structures and terrain required by the model. The concentrations reported at the sensitive receptors are not significantly affected by these uncertainties; the effect is however significant immediately beyond the property boundary.



10 Risk Assessment

LAQS used the rating system shown in Figure 33 below to attach a risk to the NAAQS being exceeded as a result of emissions from the current operations, yielding the following:

Emissions at AEL limits:

Likelihood of occurrence:

Frequency of activity: Daily, i.e. score = 5

Frequency of impact: Daily, i.e. score = 5

Confidence: High, i.e. score = 2

Total score for likelihood of occurrence: 12

Consequence:

Severity: High, i.e. score = 4

Spatial scope: Neighbouring area, i.e. score = 4

Duration: Life of operation, i.e. score = 4

Total score for consequence: 12

The overall score, i.e. likelihood x consequence, is 144 which places the potential risk in the “very high” category.

Emissions at measured concentrations:

Likelihood of occurrence:

Frequency of activity: Monthly, i.e. score = 3

Frequency of impact: Very seldom, i.e. score = 2

Confidence: High, i.e. score = 2

Total score for likelihood of occurrence: 7

Consequence:

Severity: Significant, i.e. score = 3

Spatial scope: Area specific, i.e. score = 2

Duration: Life of operation, i.e. score = 4

Total score for consequence: 9

The overall score, i.e. likelihood x consequence, is 63 which places the potential risk in the “medium-low” category.

Should abatement be applied, or stack height increased sufficiently, the likelihood of the air quality standard being exceeded becomes very low, resulting in the risk of the air quality standards being exceeded falling in the “low” category.



SPATIAL SCOPE	RATING	DURATION	RATING	SEVERITY	RATING	Confidence	RATING							
Activity Specific	1	1 day to 1 month	1	Insignificant	1	Absolute	1							
Area Specific	2	1 month to 1 year	2	Small	2	High	2							
Whole Site / Plant	3	1 year to 5 years	3	Significant	3	Moderate	3							
Neighbouring Area	4	Life of operation	4	High	4	Low	4							
Regional Area	5	Permanent	5	Disastrous	5	None	5							
FREQUENCY OF ACTIVITY	RATING	FREQUENCY OF IMPACT	RATING											
Annually or less	1	Almost never / Almost impossible	1											
6 monthly	2	Very seldom / highly unlikely	2											
Monthly	3	Infrequent / unlikely / seldom	3											
Weekly	4	Often / regularly / likely / possible	4											
Daily	5	Daily / highly likely / definitely	5											
SIGNIFICANCE RATING OF IMPACT			TIMING											
Very Low	1 to 25		Pre- Construction / prior to activity											
Low	31 to 50		Construction / installation											
Medium – Low	51 to 75		Operation / Activity											
Medium – High	76 to 100													
High	101 to 125													
Very High	126 to 150													
CONSEQUENCE (= Severity + special scope + duration)														
LIKELY HOOD (= frequency of activity + frequency of impact + confidence)														
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
3	6	9	12	15	18	21	24	27	30	33	36	39	42	45
4	8	12	16	20	24	28	32	36	40	44	48	52	56	60
5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
6	12	18	24	30	36	42	48	54	60	66	72	78	84	90
7	14	21	28	35	42	49	56	63	70	77	84	91	98	105
8	16	24	32	40	48	56	64	72	80	88	96	104	112	120
9	18	27	36	45	54	63	72	81	90	99	108	117	126	135
10	20	30	40	50	60	70	80	90	100	110	120	130	140	150

Figure 33: Impact rating system.



11 Conclusions

A conservative approach was followed in estimating emissions from the AutoX plant. The rationale behind this approach is that the outcome will be an overestimation of the impact on air quality. Should this impact prove negligible the real impact would be even less.

Three emission scenarios were assessed against that approach: the licensed emissions on its own, the actual expected plant emissions, and a cumulative assessment based on the actual emissions of the plant and surrounding sources.

On the licensed basis the predicted maximum exceeds its criterion for every pollutant and averaging period assessed. On the measured and expected basis, the model shows that the annual average lead concentration is likely to exceed the national ambient air quality standard, but that the exceedance is confined to the immediate vicinity of the plant.

The issue remains that the licence and impact assessment must consider the fact that an operator is allowed to emit up to the maximum permitted concentration and the applied flow rate. It is therefore critical that the information contained in the licence is accurate and relevant in terms of the process classifications, controlled pollutants, and flow rates.

To reduce the predicted air quality impact the emissions, allowed for in the licence, must either be reduced or the effective release height of the emissions must be increased to allow for better dispersion before the pollutants reach the ground.

To reduce total emissions from the site, it is recommended that the listed stacks, and their associated processes, are carefully evaluated to determine the applicability of the Listed Activity to those processes. Processes that do not form part of the listed activity should be excluded from the AEL.

Additionally, the maximum allowed emissions may be reduced by the Licensing Authority, who is duly authorised to do so by the NEMA: Air Quality Act, via the introduction of stricter emission limits than the nationally listed emission limits stated in GN 893.

Alternatively, dispersion can be improved by increasing the effective release height of the pollutants, thus allowing more time for dispersion before the pollutants reach ground-level. A repeated dispersion modelling study will be required to obtain the minimum stack heights required to prevent exceedance of the NAAQS levels.

LAQS recommends that a revised AEL is issued to AutoX to allow the company to investigate options to reduce the ground-level concentrations of lead. It is also recommended that AutoX considers an ambient dust monitoring program, particularly in the easterly corner of the property where the highest ground-level concentrations are expected. The monitoring program should be targeted to the collection of airborne dust samples and subsequent analysis of lead content for comparison against NAAQS levels.



APPENDIX A List of sources

This appendix lists the point sources of AutoX included in the dispersion model, with the release parameters and emission rates as modelled.

Coordinates are given in decimal degrees on WGS84, as they appear in the licence application. Emission rates are in grams per second. NO_x is modelled as emitted and converted to NO₂ for assessment at the ratio given in Section 5.2.3.

Scenario B and C uses the same plant on the measured basis. The 2025 stack survey revises the exhaust temperature, exit velocity and internal diameter as well as the emission rate, so the release parameters below differ from Table A1 and not only the rates. AX4 stands on the licence but has since been physically replaced and is not included in Scenario B or C.



Source ID	Description	Coordinates (lat, lon)	Stack height (m)	Diameter (m)	Exit temp. (K)	Exit velocity (m/s)	PM ₁₀ (g/s)	Pb (g/s)	SO ₂ (g/s)	NOx (g/s)	H ₂ SO ₄ (g/s)
AX1	TBS Assembly	-33.926190, 25.573754	4	0.9	293.1	10.9	0.1939	0.0129	—	—	—
AX2	Charge Room Extraction Stack 1	-33.926456, 25.573544	8	1.2	302.7	15.5	—	—	5.5369	5.5369	0.4845
AX3	Charge Room Extraction Stack 2	-33.927156, 25.572949	8	1.2	309.5	16.6	—	—	5.7990	5.7990	0.5074
AX4	BM Assembly Extraction 4	-33.926294, 25.574358	6	1	303.1	9	0.1911	0.0127	—	—	—
AX6	BM Assembly Extraction 1	-33.926357, 25.574455	6	1	297.1	12.4	0.2687	0.0179	—	—	—
AX7	BM Assembly Extraction 2	-33.926435, 25.574591	6	1	299.1	12.43	0.2675	0.0178	—	—	—
AX8	BM Assembly Extraction 3	-33.926509, 25.574708	6	1	298.4	8.64	0.1863	0.0124	—	—	—
AX9	Plate Making Extraction	-33.927088, 25.575646	5	0.9	309.9	31.19	0.5247	0.0350	—	—	—
AX10	Lead Oxide Manufacturing Extraction	-33.927383, 25.575110	6.5	1.2	304.3	5.84	0.1780	0.0119	—	—	—
AX12	Oxide Mill Silo Stack (Negative Oxide)	-33.927414, 25.575396	5.5	0.4	347.4	7.43	0.0220	0.0015	—	—	—
AX13	Oxide Mill Silo Stack (Positive Oxide)	-33.927456, 25.575349	5.5	0.5	372.0	5.32	0.0230	0.0015	—	—	—
AX14	Post-Classifier Bag Filters 1, 2 and 3	-33.926561, 25.574815	8.5	0.25	305.4	7.12	0.0094	0.0006	—	—	—
AX15	Post-Classifier Bag Filters 4,5 and 6	-33.927256, 25.575409	7.75	0.25	319.6	12.34	0.0155	0.0010	—	—	—
AX16	Paste Mixer Stack	-33.927236, 25.575371	6	0.5352	298.6	3.17	0.0196	0.0013	—	—	—



Source ID	Description	Coordinates (lat, lon)	Stack height (m)	Diameter (m)	Exit temp. (K)	Exit velocity (m/s)	PM ₁₀ (g/s)	Pb (g/s)	SO ₂ (g/s)	NOx (g/s)	H ₂ SO ₄ (g/s)
AX17A	Curing Oven Electric	-33.926896, 25.574992	6.5	0.3385	322.9	14.74	0.0337	0.0022	—	—	—
AX17B	Curing Oven Electric	-33.927113, 25.574962	6.5	0.3385	322.9	14.74	0.0337	0.0022	—	—	—
AX17C	Curing Oven Electric	-33.927140, 25.574937	6.5	0.3385	322.9	14.74	0.0337	0.0022	—	—	—
AX17D	Curing Oven Electric	-33.927164, 25.574916	6.5	0.3385	322.9	14.74	0.0337	0.0022	—	—	—
AX17E	Curing Oven Electric	-33.927190, 25.574894	6.5	0.3385	322.9	14.74	0.0337	0.0022	—	—	—
AX17F	Curing Oven Electric	-33.926940, 25.574834	6.5	0.3385	322.9	14.74	0.0337	0.0022	—	—	—
AX18A	Curing Oven Gas Fired	-33.926771, 25.574975	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18B	Curing Oven Gas Fired	-33.926792, 25.574956	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18C	Curing Oven Gas Fired	-33.926809, 25.574941	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18D	Curing Oven Gas Fired	-33.926829, 25.574924	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18E	Curing Oven Gas Fired	-33.926845, 25.574910	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18F	Curing Oven Gas Fired	-33.926866, 25.574891	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18G	Curing Oven Gas Fired	-33.926884, 25.574874	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX18H	Curing Oven Gas Fired	-33.926906, 25.574856	7	0.35	319.5	1.77	0.0044	0.0003	—	—	—
AX19	La Pneumatic Stack (Lid Sealing)	-33.926792, 25.574956	7.5	0.4	307.7	3.41	0.0114	0.0008	—	—	—
AX20A	Battery Acid Filling	-33.926520, 25.573880	5	0.45	296.0	10.3	—	—	0.5290	0.5290	0.0463



Source ID	Description	Coordinates (lat, lon)	Stack height (m)	Diameter (m)	Exit temp. (K)	Exit velocity (m/s)	PM ₁₀ (g/s)	Pb (g/s)	SO ₂ (g/s)	NOx (g/s)	H ₂ SO ₄ (g/s)
AX20B	Battery Acid Filling	-33.926520, 25.573880	5	0.45	296.0	10.3	—	—	0.5290	0.5290	0.0463
AX20C	Battery Acid Filling	-33.926520, 25.573880	5	0.45	296.0	10.3	—	—	0.5290	0.5290	0.0463
AX20D	Battery Acid Filling	-33.926520, 25.573880	5	0.45	296.0	10.3	—	—	0.5290	0.5290	0.0463
AX22	BM Assembly Extraction 4	-33.926618, 25.573601	4	0.25	298.2	25.05	0.0338	0.0023	—	—	—
AX23	Proposed Expansion Stack 1	-33.926652, 25.573568	6	0.55	328.2	13.78	0.0817	0.0054	—	—	—
AX24	Proposed Expansion Stack 2	-33.926660, 25.573676	4	0.55	306.1	22.02	0.1401	0.0093	—	—	—
AX25	Proposed Expansion Stack 3	-33.926707, 25.573632	6	0.25	298.2	25.05	0.0338	0.0023	—	—	—
AX26	Proposed Expansion Stack 4	-33.926440, 25.572403	6	0.25	298.2	25.05	0.0338	0.0023	—	—	—
AX27	Proposed Expansion Stack 5	-33.926404, 25.572380	6	0.25	298.2	25.05	0.0338	0.0023	—	—	—
AX28	Proposed Expansion Oven Stack	-33.926398, 25.572434	6	0.25	395.1	5.91	0.0100	—	0.1003	0.1003	—
Total — 40 sources							2.5152	0.1670	13.5522	13.5522	1.1771

Table A1: On-site sources — AutoX, modelled in Scenario A (AEL application flow rates at the maximum permitted emission concentrations).



Source ID	Description	Coordinates (lat, lon)	Stack height (m)	Diameter (m)	Exit temp. (K)	Exit velocity (m/s)	PM ₁₀ (g/s)	Pb (g/s)	SO ₂ (g/s)	NOx (g/s)	H ₂ SO ₄ (g/s)
AX1	TBS Assembly	-33.926190, 25.573754	4	0.9	296.2	38.08	0.0203	0.0030	—	—	—
AX2	Charge Room Extraction Stack 1	-33.926456, 25.573544	8	1.2	299.4	12.12	—	—	—	—	0.0085
AX3	Charge Room Extraction Stack 2	-33.927156, 25.572949	8	1.2	289.5	13.39	—	—	0.0001	—	—
AX4	BM Assembly Extraction 4	-33.926294, 25.574358	6	1	303.1	9	0.1911	0.0127	—	—	—
AX6	BM Assembly Extraction 1	-33.926357, 25.574455	6	1	305.9	26.08	0.0254	0.0010	—	—	—
AX7	BM Assembly Extraction 2	-33.926435, 25.574591	6	1	296.9	5.53	0.0019	0.0001	—	—	—
AX8	BM Assembly Extraction 3	-33.926509, 25.574708	6	1	302.0	22.22	0.0498	0.0004	—	—	—
AX9	Plate Making Extraction	-33.927088, 25.575646	5	0.9	307.8	7.59	0.0433	0.0007	—	—	—
AX10	Lead Oxide Manufacturing Extraction	-33.927383, 25.575110	6.5	1.2	321.4	3.03	0.0026	0.0054	—	—	—
AX12	Oxide Mill Silo Stack (Negative Oxide)	-33.927414, 25.575396	5.5	0.4	322.4	5.46	0.0051	0.0046	—	—	—
AX13	Oxide Mill Silo Stack (Positive Oxide)	-33.927456, 25.575349	5.5	0.5	378.9	9.85	0.0095	0.0015	—	—	—
AX14	Post-Classifer Bag Filters 1, 2 and 3	-33.926561, 25.574815	8.5	0.4	326.6	3.27	0.0005	0.0002	—	—	—
AX15	Post-Classifer Bag Filters 4,5 and 6	-33.927256, 25.575409	7.75	0.3	380.0	4.77	0.0002	0.0001	—	—	—



Source ID	Description	Coordinates (lat, lon)	Stack height (m)	Diameter (m)	Exit temp. (K)	Exit velocity (m/s)	PM ₁₀ (g/s)	Pb (g/s)	SO ₂ (g/s)	NO _x (g/s)	H ₂ SO ₄ (g/s)
AX16	Paste Mixer Stack	-33.927236, 25.575371	6	0.5411	299.0	7.78	0.0034	0.0006	—	—	—
AX17A	Curing Oven Electric	-33.926896, 25.574992	6.5	0.3	346.1	1.76	0.0005	0.0001	—	—	—
AX17B	Curing Oven Electric	-33.927113, 25.574962	6.5	0.3	346.1	1.76	0.0005	0.0001	—	—	—
AX17C	Curing Oven Electric	-33.927140, 25.574937	6.5	0.3	346.1	1.76	0.0005	0.0001	—	—	—
AX17D	Curing Oven Electric	-33.927164, 25.574916	6.5	0.3	346.1	1.76	0.0005	0.0001	—	—	—
AX17E	Curing Oven Electric	-33.927190, 25.574894	6.5	0.3	346.1	1.76	0.0005	0.0001	—	—	—
AX17F	Curing Oven Electric	-33.926940, 25.574834	6.5	0.3	346.1	1.76	0.0005	0.0001	—	—	—
AX18A	Curing Oven Gas Fired	-33.926771, 25.574975	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18B	Curing Oven Gas Fired	-33.926792, 25.574956	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18C	Curing Oven Gas Fired	-33.926809, 25.574941	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18D	Curing Oven Gas Fired	-33.926829, 25.574924	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18E	Curing Oven Gas Fired	-33.926845, 25.574910	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18F	Curing Oven Gas Fired	-33.926866, 25.574891	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18G	Curing Oven Gas Fired	-33.926884, 25.574874	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX18H	Curing Oven Gas Fired	-33.926906, 25.574856	7	0.3385	304.7	3.12	0.0008	0.0002	—	—	—
AX19	La Pneumatic Stack Sealing)	-33.926792, 25.574956	7.5	0.4	303.5	2.59	0.0006	0.0002	—	—	—



Source ID	Description	Coordinates (lat, lon)	Stack height (m)	Diameter (m)	Exit temp. (K)	Exit velocity (m/s)	PM ₁₀ (g/s)	Pb (g/s)	SO ₂ (g/s)	NOx (g/s)	H ₂ SO ₄ (g/s)
AX20A	Battery Acid Filling	-33.926520, 25.573880	5	0.45	305.9	7.81	—	—	0.0047	0.0012	<0.0001
AX20B	Battery Acid Filling	-33.926520, 25.573880	5	0.45	305.9	7.81	—	—	0.0047	0.0012	<0.0001
AX20C	Battery Acid Filling	-33.926520, 25.573880	5	0.45	305.9	7.81	—	—	0.0047	0.0012	<0.0001
AX20D	Battery Acid Filling	-33.926520, 25.573880	5	0.45	305.9	7.81	—	—	0.0047	0.0012	<0.0001
AX22	BM Assembly Extraction 4	-33.926618, 25.573601	4	0.3	305.2	22.98	0.0012	0.0001	—	—	—
AX23	Proposed Expansion Stack 1	-33.926652, 25.573568	6	0.55	328.2	13.78	0.0050	0.0001	—	—	—
AX24	Proposed Expansion Stack 2	-33.926660, 25.573676	4	0.55	306.1	22.02	0.0085	0.0002	—	—	—
AX25	Proposed Expansion Stack 3	-33.926707, 25.573632	6	0.25	298.2	25.05	0.0020	<0.0001	—	—	—
AX26	Proposed Expansion Stack 4	-33.926440, 25.572403	6	0.25	298.2	25.05	0.0020	<0.0001	—	—	—
AX27	Proposed Expansion Stack 5	-33.926404, 25.572380	6	0.25	298.2	25.05	0.0020	<0.0001	—	—	—
AX28	Proposed Expansion Oven Stack	-33.926398, 25.572434	6	0.25	395.1	5.91	0.0007	0.0001	<0.0001	—	—
Total — 39 sources							0.1936	0.0201	0.0189	0.0049	0.0086

Table A2: On-site sources — AutoX, modelled in Scenario B and C (2025 stack survey; unmeasured and proposed stacks at their maximum permitted rates).